Estimating the Sign and Magnitude of the Inflation Risk Premium

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Abstract

Beginning in early 2021, U. S. financial markets and policy makers began to recognize a rising rate of inflation, caused by the prevailing health crisis, provisions of monetary and fiscal Federal liquidity and supply-chain disruptions. In addition to the prevailing rate of inflation, one of the acute policy issues that arose was the projection of future inflation rates. In this context, financial markets provide two forward-looking metrics for U. S. inflation: The break-even rate of inflation (the difference between nominal and TIPS yields for a given maturity), and the rates in inflation swap contracts.

The paper's key contributions can be summarized as follows: First, using both realized inflation and survey-based expectations as benchmarks, we document negative inflation risk premia that are strong and robust, especially at short horizons, corroborated across several measures. This pattern holds across multiple unconditional measures. Second, we propose a simple CAPM-based model that explains why negative inflation risk premia can arise and shows that the systematic risk embedded in breakeven inflation and inflation swap prices is informative about inflation risk premia. Third, we provide a simple regression-based test that highlights maturity-related features of the TIPS market around 2000 and documents the presence of lingering illiquidity. Despite these issues, our inflation risk premium estimates are robust, as the key properties we identify also hold in the inflation swap market.

Key Words: TIPS securities; Inflation Risk Premium

JEL Classification: E31 – Inflation; G12 – Asset Pricing; G13 – Contingent Pricing

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1 Introduction

At this time in economic history, it is difficult to overstate the importance of forecasting the rate of inflation. Subsequent to the responses of fiscal and monetary authorities to the health crisis of 2020, inflation was said to have "reared its ugly head," requiring a set of responses by the Federal Reserve Bank Federal Open Market Committee that began in March 2022, and whose future path is uncertain as of this writing.

In terms of the data regulators as well as other economists have at their disposal, the retrospective realized rates of inflation published by the Labor Dept.'s Bureau of Labor Statistics is arguably the least important. Rather, we have at our availability three measures of prospective inflation:

1. Published on a quarterly frequency, the Federal Reserve Bank of Philadelphia publishes the "Short-Term and Long-Term Inflation Forecasts: Survey of Professional Forecasters." Assuming good faith on the parts of the survey participants, the survey reflects the responders' assessment of inflation.

2. For maturities from one to thirty years, financial markets provide two important, forward-looking assessments of inflation:

   (a) The Break-Even Rate of Inflation (hereafter BE), typically computed as the difference between the U. S. Treasury's on-the-run (most recently issued) par rate of interest and the corresponding rate on the Treasury's TIPS ("Treasury Inflation Protected Securities") instruments. [1]

   (b) The rate of inflation embedded in Inflation Swap contracts (hereafter IS)

[1] The Federal Reserve Bank of St. Louis does a commendable job of providing a time-series of the five-year break even rate of inflation on its `https://fred.stlouisfed.org/series/T5YIE/` website.

As these are a consequence of market activity, the data availability here is daily. The U. S. Treasury issued its first TIPS security in Jan. 1997, but the methodical collection of this data lags somewhat.[2]

It is well-known from other financial markets, these markets embed financial risk premiums. The consequence of these risk premiums is financial markets "are under no obligation" to provide statistically unbiased estimates of the future realization of a particular instrument or metric. Thus, the equity risk premium implies the expected value of a zero-dividend share of stock is not the current value compounded at the riskfree rate. Similarly, and more closely to what we consider here, a A possibly non-zero oil-price risk premium implies the current price of a distant oil futures contract is not an unbiased estimate of the spot price at that future date. Finally, the well-documented case of a volatility risk premium implies the VIX implied-volatility index is likely an upward-biased predictor of the path of realized volatility for the S&P 500. Finally, most directly for our purposes, a non-zero inflation risk premium implies that the break-even and inflation-swap rates are not unbiased estimates of expected inflation

Before proceeding further, it is important to lay out three specific issues which need be addressed in the estimation of the inflation risk premium and which any empirical assessment must confront:

1. The first has already been alluded to above — the matter of possible differential liquidity between the TIPS and the traditional nominal bonds. Naturally, any empirical determination on the difference between the daily values of forward-looking inflation rates, BE or IS, must be tempered by the issue of whether one or both is contaminated by the issue of differential liquidity of TIPS and nominal Treasury securities.

[2]As discussed in detail below, it is entirely legitimate to raise the question of whether TIPS are subject to an illiquidity premium relative to their nominal, on-the-run Treasury counterparts. A closely related issue is to attempt to identify the date at which the TIPS market "matured."

For an interesting timeline of TIPS issuance, see the Treasury's publication, `https://www.treasurydirect.gov/research-center/timeline/tips/`.

2. The second issue pertains to a direct measurement of the inflation risk premium: The difference between BE or IS and either realized inflation or survey forecasts of inflation. In this regard, the use of realized inflation as one measure of the inflation risk premium is predicated on the predicate of efficient markets, under which the market will deliver the ex-ante expectation "on average" (over sufficiently long time-series to determine statistical significance).[3]

3. The third item pertains to the matter of whether any systematic risk embedded in the values of the BE or IS values is reflective of, and therefore needs be accounted for, in the computation of the inflation risk premium.

   In quantifying the magnitude of the systematic risk premium, we choose as our driving model the Capital Asset Pricing Model. Notwithstanding the development of multi-factor models, we note our approach is grounded in the foundational logic of systematic risk pricing. Whereas the market return in the CAPM is a well-established systematic risk factor, the status of other factors — such as SMB and HML, to say nothing of Momentum — remains debated, particularly regarding whether they represent true priced risks or empirical anomalies. This is even more the case for level, slope, and curvature. These are not systematic risk factors in the CAPM sense, but rather state variables in affine term structure models. While yield curve factors are useful for modeling the dynamics of the term structure, they are not priced risk factors in the sense of CAPM or APT.

   By focusing on the market risk premium, our model offers a tractable benchmark for understanding the inflation risk premium within a well-established theoretical framework.

Especially in light of their importance for forecasting inflation, as well as the

[3]Based on rational expectations, we use realized inflation as an ex-post proxy for investors' ex-ante expectation. We nonetheless acknowledge that, in any finite sample, realized inflation can diverge from expectations because of inflation shocks rather than risk premia, and we accordingly interpret the realized-inflation IRPs with caution.

attendant investment and policy implications, this research seeks to understand whether these measures of prospective inflation contain an inflation-risk premium, and what the sign and magnitude of that inflation risk premium are. Perhaps surprisingly, based on the multiple empirical approaches we implement, we document survey- and realization-based inflation risk premia that are negative, especially at short horizons: Market-based inflation is lower than expected inflation, with the gap narrowing as the maturity lengthens.

To provide a theoretical rationale for the presence of negative inflation risk premia, we propose a CAPM-based framework that decomposes the premium into three structural components: The market price of risk (Sharpe Ratio), the quantity of risk (inflation volatility), and the systematic nature of risk (correlation with the market). While the former two determine the magnitude, the sign of the risk premia is exclusively governed by the correlation between inflation and market returns. We document a persistently positive correlation in inflation-related assets (inflation swap and break-even rates of inflation), implying inflation is perceived to be pro-cyclical by market participants. Consequently, nominal Treasury bonds function as effective hedging assets, preserving real value during disinflationary market downturns. This insurance property induces investors to accept break-even rates systematically below expected inflation, thereby rationalizing the negative inflation risk premia observed in the data.

The paper's key contributions can be summarized as follows:

1. We document a set of simple survey- and realization-based inflation risk premia that are robustly negative most strongly at short horizons and fading toward zero at the long horizon:: Market-based inflation is lower than expected inflation, especially in the near term.

2. To augment those tests, we propose a simple CAPM-based framework that

   (a) Provides an intuitive and disciplined explanation for why negative inflation risk premia can arise,

   (b) Links inflation risk premia to systematic risk embedded in break-even rates and inflation swaps, and

(c) Delivers a risk-adjusted measure of inflation expectations that closely tracks survey expectations.

3. We propose a simple regression-based test that addresses maturity and liquidity concerns in the early TIPS market. Importantly, we show that our inflation risk premium estimates are robust to these issues, as the same negative Inflation Risk Premium (IRP) patterns are consistently observed in the inflation swap market as well.

This paper is now structured as follows. Since we are by no means the first to consider the estimation of the inflation risk premium, Section 2 delves in some detail into the broad extant literature that has previously addressed this matter, and we take care to decompose our consideration of the literature by relating our work to each of the three critical sub-issues we have just identified. Section 3 will briefly describe the data, and its sources, we used in this research. In turn, Section 4 will present our empirical works addressing each of these three issues. Section 5 will take a slight detour into a public-policy implication of the markets for TIPS and inflation swaps. Section 6 concludes.

2 Literature Review

It is entirely unsurprising the literature on index-linked instruments is wide and extensive. To do justice to that previous research, we relegate the predominant presentation of the literature review to Appendix A, restricting our analysis here to an overview of the literature, and to distinguishing our work from previous research. To do so, we present the previous research in three different sub-categories:

2.1 Measuring the Inflation Risk Premium

A number of studies have emphasized the role of inflation risk premia, which have been interpreted as tending to push break-evens above true expected inflation. Haubrich, Pennacchi and Ritchken (2012) show that inflation risk premia embedded in inflation swaps and TIPS can generate an upward bias of roughly 10 – 40 basis points, with the magnitude varying across horizons. Campbell, Shiller and Viceira (2009) similarly argue that investors demand compensation for inflation

uncertainty, causing break-evens to exceed survey-based expectations, particularly at longer maturities. Hördahl and Tristani (2012) provide complementary evidence for the euro area, estimating positive inflation risk premia of 20 – 30 basis points, which push break-evens above model-implied expectations. Sarte (2012) adds a macro-structural perspective, showing that risk-premium variation tied to business-cycle conditions can generate substantial time-variation in the upward bias of break-evens.

A growing literature examines how monetary policy regimes and communication shape break-even biases. Casiraghi and Miccoli (2019) show that central-bank credibility and forward-guidance shocks can alter the balance between liquidity and risk premia, generating shifts in break-even bias of 10 - 25 basis points around major policy events. Binder and Kamdar (2022) document that disagreement and uncertainty in survey-based expectations can amplify the risk-premium component of break-evens, increasing upward bias during periods of elevated macro uncertainty.

Taken together, the literature demonstrates that while break-even inflation rates contain valuable information about market-based inflation expectations, they are not unbiased predictors of future inflation. Instead, break-evens reflect a combination of liquidity premia, risk premia, market segmentation and occasional mispricing, each of which can materially distort the signal. As a result, empirical work that relies on break-evens must adjust for these premia or interpret break-evens as inflation compensation rather than pure expectations.

Our contribution in this realm is to provide the unconditional estimate of four risk premium estimates calculated over the years. In contrast to the prevailing view that risk premia push break-evens above expected inflation, these measures point to a negative premium with an upward-sloping term structure, most pronounced at short horizons and approaching zero or positive at the longest.

2.2 The Impact of Liquidity Risk

A large literature examines whether TIPS break-even inflation rates provide unbiased forecasts of future inflation, and the consensus is that break-evens embed several premia that generate systematic and time-varying biases. Early research highlights the role of liquidity distortions in the TIPS market, particularly in the

early 2000s when the market was thin and trading conditions uneven. Durham (2006) provides some of the earliest evidence that TIPS liquidity premia can meaningfully depress break-evens, with estimated downward biases of 20 – 50 basis points. Gürkaynak, Sack, and Wright (2010) similarly show that limited liquidity and market segmentation produced a downward bias in break-evens, typically in the range of 20 – 40 basis points, with the bias widening during periods of financial stress. Christensen and Gillan (2012) further document that negative liquidity premia can depress break-evens by 30 – 60 basis points, reinforcing the view that raw break-evens systematically understate expected inflation when TIPS liquidity deteriorates.

Shen and Corning (2001) argued that the yield difference between nominal Treasuries and TIPS has not been a reliable measure of market inflation expectations due to a large and variable liquidity premium on TIPS. However, they suggest that as liquidity conditions in the two Treasury markets converge in the future, the yield difference could become a better measure of inflation expectations. Shen (2006) points out that attributing changes in yield spreads solely to changes in market inflation expectations while ignoring the liquidity risk premium could lead to an overstatement of inflation expectations, while conceding liquidity risk premia have likely declined in recent years. Christensen and Gillan (2011) use information from inflation swaps to estimate the possible range of the liquidity premium in TIPS, suggesting the estimates of inflation expectations derived from TIPS, after accounting for the liquidity premium, are well below the inflation forecasts provided by the Survey of Professional Forecasters. Driessen, Nijman, and Simon (2017) show that the level of liquidity significantly impacts TIPS prices, while swap yields include a liquidity risk premium. Andreasen, Christensen, and Riddell (2021): This paper introduces a novel model to identify liquidity risk from individual TIPS prices, taking into account the tendency of TIPS to be held by buy-and-hold investors.

While most researchers find liquidity premiums have declined over time, Zeng (2013) finds the sharp declines in 10-year and 5-year break-even inflation rates in 2009 were driven by a substantial increase in liquidity risk rather than a decrease in inflation expectations.

A related strand of work focuses on relative mispricing between TIPS and nom-

inal Treasuries. Fleckenstein, Longstaff and Lustig (2014) identify persistent arbitrage violations between the two markets, implying that TIPS are often undervalued relative to their nominal counterparts. Their estimates suggest that such mispricing can depress break-evens by 30 – 80 basis points, depending on maturity and market conditions. Pericoli (2012) extends this analysis to European inflation-linked markets, showing that liquidity and segmentation effects similarly distort euro-area break-evens, often producing downward biases of 20 – 40 basis points. Together, these studies strengthen the argument that break-evens cannot be interpreted as unbiased measures of expected inflation without correcting for structural pricing distortions.

More recent work integrates these competing forces using structural decompositions and high-frequency identification. D'Amico, Kim and Wei (2018) develop a joint model of nominal and real yields that separates expected inflation, liquidity premia and inflation risk premia. Their estimates show that liquidity premia typically depress break-evens by 20 – 50 basis points, while risk premia raise them by 10 – 30 basis points, producing a net bias that ranges from -40 to $+30$ basis points depending on market conditions. Bauer and McCarthy (2015) use high-frequency yield curve movements around macro announcements to identify the inflation-expectations component of break-evens, finding that risk premia can account for 15 – 35 basis points of upward bias. Chen, Liu and Cheng (2010) similarly show that inflation-linked markets in Asia exhibit both liquidity-driven downward biases and risk premium-driven upward biases, with magnitudes comparable to those in the U. S. and Europe.

Our contribution here is a simple regression-based test that dates the maturing of the TIPS market to 2000–2001 and documents its lingering illiquidity, while showing this does not drive our central finding, since the same negative premia hold in the inflation-swap market.

A general finding from the aforementioned studies is the positive inflation risk premia that push break-evens above expected inflation. Our finding that inflation risk premia are predominantly negative, most pronounced at short horizons, reflects differences in sample period, identification and data construction relative to earlier studies. First, our analysis begins in late 2006, thereby excluding the early TIPS era

and the large episodes of TIPS undervaluation documented by Haubrich, Pennacchi and Ritchken (2012), which are a primary source of the positive premia reported in earlier work. Second, we anchor expected inflation using Survey of Professional Forecasters data at a quarterly frequency, rather than inferring expectations from breakevens or model-implied paths; this survey-based benchmark removes much of the upward bias in breakevens emphasized in Campbell, Shiller and Viceira (2009) and Hrdahl and Tristani (2012). Third, we address maturity and liquidity concerns in the early TIPS market with a simple regression-based test and show that the same negative inflation risk premium patterns appear in the inflation swap market, confirming that our results are not driven by TIPS-specific liquidity distortions. Finally, our horizon definitions and post-GFC sample place more weight on a regime in which inflation uncertainty is low and survey-anchored term-structure models — including Federal Reserve staff work — also find small or negative inflation risk premia.[4] Once these differences are accounted for, our results are fully consistent with the more recent, survey-anchored literature and should be interpreted as a characterization of the post-2006 regime rather than a contradiction of earlier, pre-2006 findings.

3 Data

3.1 Overview of Data

This study required the collection of data from several sources:

1. The Break-Even Rates of Inflation for maturities of 1, 2, 3, 5, 10 and 30 yrs.: Bloomberg L.P. As is well-known, these are the differences between the nominal yields to maturity and those of TIPS securities

2. Zero-coupon Inflation Swap Rates for maturities of 1, 2, 3, 5, 10 and 30 yrs.: Bloomberg L.P.

[4]For those confirming analyses, see Haubrich, Pennacchi and Ritchken (Cleveland Fed working paper + RFS 2012), D'Amico, Kim and Wei (2018 and related Fed term-structure work).

3. Implied Inflation Volatilities: Taken from at-the-money inflation options provided by Bloomberg L.P. using the Shifted LogNormal model. The series is obtained at a quarterly frequency aligned with the deadline dates of each survey round of the SPF.

4. Inflation Forecasts and Measures of Dispersion: Survey of Professional Forecasters

5. Realized values of the Consumer Price Index for All Urban Consumers (CPI-U): Bureau of Labor Statistics

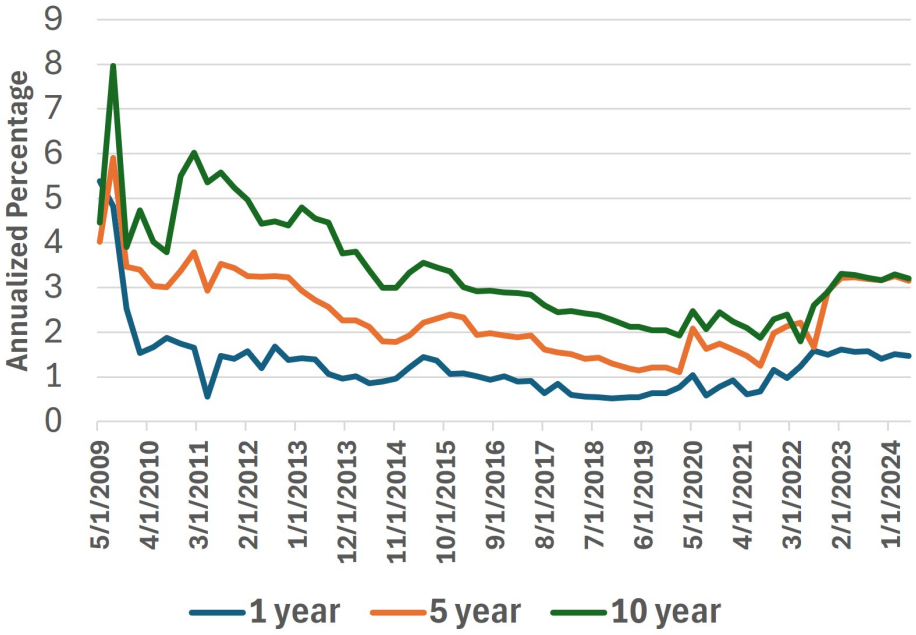
Due to data availability on Bloomberg, the sample period runs from November 10, 2006, to October 26, 2022. The starting period was chosen as it marks the date from which the time series of Break-Even Rates (BE) and Inflation Swap Rates (IS) were stable across all maturities. For the analysis that involves the use of implied volatilities the sample period runs from May 12, 2009, to August 29, 2022. The starting period was chosen as it marks the date when the time series of implied volatility of inflation from Bloomberg became available. For the analysis conducted at the quarterly frequency, the sample period runs from 2009 Q2 – 2022 Q3.

3.2 Implied Inflation Volatilities

Figure 1 compares the time series of implied volatilities of inflation and the dispersion of inflation forecasts from the SPF across three maturities: 1, 5, and 10 years. There is a notable difference between the two measures. While the implied volatility series exhibits an upward-sloping term structure, the forecast dispersion displays a downward-sloping term structure. In other words, while the market's measure of inflation uncertainty increases with the horizon, the survey-based measure of inflation uncertainty decreases. The former may reflect the market's perceived risk whereas the latter may be due to forecasters agreeing on the stable, long-term trend of inflation.

(PLACE FIG. 1 HERE)

Implied Inflation Volatilities



Dispersion in Inflation Forecasts

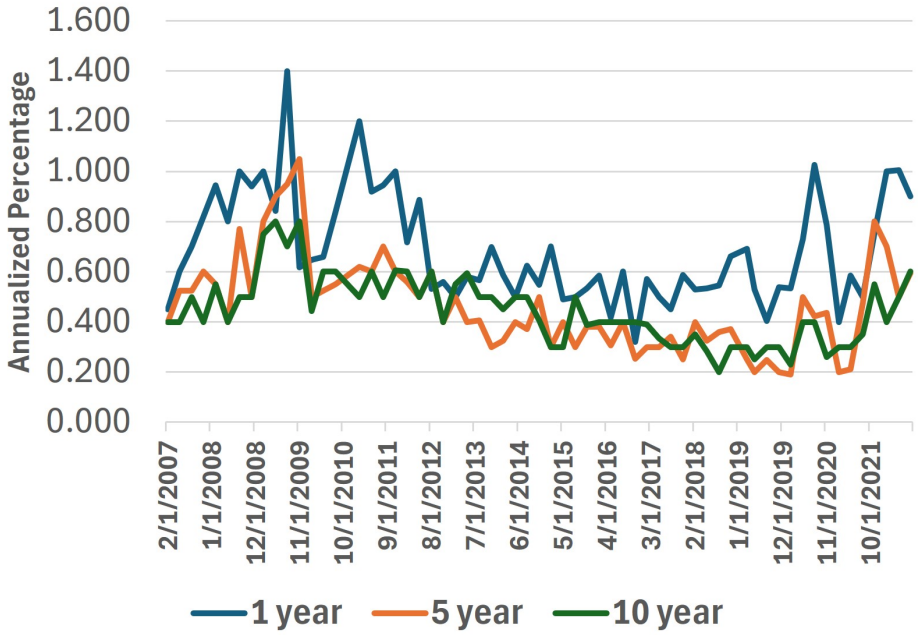


Table 1 presents the correlation matrix of forecast dispersion and ZC volatilities. Both series are positively correlated and it gets stronger over the horizon. The two series share several similarities. They are negatively correlated with market return and positively correlated with market uncertainty both in terms of historical volatility ("SPX Vol") and implied volatility (VIX). However, while survey dispersion tends to increase with trend inflation (Inflation Y/Y), implied volatility tends to decrease with trend inflation.

Table 1: Correlation Matrix of Forecast Dispersion and Implied Volatilities of Inflation

	ZC Vol (1 yr.)	ZC Vol (5 yr.)	ZC Vol (10 yr.)	SPX Return	SPX Vol	VIX	Inflation Y/Y
Dispersion (1 yr.)	0.460			−0.199	0.367	0.457	0.191
Dispersion (5 yr.)		0.738		−0.186	0.394	0.440	0.110
Dispersion (10 yr.)			0.773	−0.074	0.293	0.280	−0.023
ZC Vol (1 yr.)	1.000			−0.143	0.342	0.342	−0.315
ZC Vol (5 yr.)		1.000		−0.180	0.245	0.344	−0.261
ZC Vol (10 yr.)			1.000	−0.171	0.222	0.306	−0.264

Notes:

1. This table presents the correlation matrices of inflation forecast dispersion from the SPF and the implied volatility from the Bloomberg with different data.

2. SPX Return refers to the market return computed using S&P 500 index. SPX Vol refers to the historical volatility of S&P 500 index. VIX is the CBOE Volatility Index.

3. The sample period runs from 2009Q2 to 2022Q3, with dates aligned based on the SPF survey deadline.

4 Estimating the Unconditional Inflation Risk Premium

Unconditionally, and ignoring aspects of systematic risk, the issue of establishing the sign of the inflation risk premium (IRP) would appear disarmingly simple. Consider:

Assume investors are averse to inflation uncertainty. If so, assume they prefer a known real return to an expected real return of the same value:

$$E\,(\text{ Real Return on Nominal Bond }) > \text{Real Rate on TIPS} \tag{1}$$

$$\text{Nominal Return} - E\,(\text{ Inflation }) > \text{Real Rate on TIPS} \tag{2}$$

Rearranging eq. (2) yields

$$E\,(\text{ Inflation }) < \text{Nominal Return} - \text{Real Rate on TIPS} = \text{Break-Even Inflation} \tag{3}$$

Per eq. (3), a risk-averse investor would require positive inflation risk premium meaning that the break-even inflation (or inflation swap) would be higher than the expected inflation. However, the data "begs to differ": To illustrate this, we define IRPs with maturity h as follows.

$$\begin{aligned}
\text{IRP}_{1,h} &= \text{BE}_{t,h} - \pi_{t,t+h,\text{arithmetic}} \\
\text{IRP}_{2,h} &= \text{BE}_{t,h} - E_t\left(\pi_{t+h}\right) \\
\text{IRP}_{3,h} &= \text{IS}_{t,h} - \pi_{t,t+h,\text{geometric}} \\
\text{IRP}_{4,h} &= \text{IS}_{t,h} - E_t\left(\pi_{t+h}\right)
\end{aligned}$$

$E_t\left(\pi_{t+h}\right)$ denotes h-year ahead inflation expectation from the SPF survey data. $\pi_{t,t+h,\text{arithmetic}}$ and $\pi_{t,t+h,\text{geometric}}$ refer to h–year ahead realized inflation, annualized using arithmetic and geometric formulas, respectively, to account for the different annualization structures of break-even inflation and inflation swaps.

To compute the realized inflation, we computed the reference CPI following Gürkaynak, Sack and Wright (2010). After computing the percentage change in the reference CPI levels, both were annualized using both arithmetic and geometric average.

To obtain expected inflation, the Survey of Professional Forecasters (SPF) provides 1-year, 5-year, and 10-year ahead forecasts.

- To align the timing of the survey with the daily swap and break-even inflation, the *deadline* of each survey served as the benchmark date.
- To consider the indexation lag, we reconstructed the 1-year ahead forecast so that 1-year spans from the previous quarter to three quarters later. The reconstruction was possible as the survey provides microdata on quarterly backcast, nowcast[5] and up to four-quarter ahead forecast.[6]
- 5-year and 10-year inflation expectations are changes in price level from fourth quarter of current year to the fourth quarter five or ten years later. We used those directly without any adjustments.

The IRPs are presented in Figure 2. The x-axis represents maturity and y-axis is the average IRP in percentage points. Average IRP are shown with 95% confidence band using Newey-West Standard Errors. Whether we use survey expectation or realized data, there is a *negative* IRP in the nearest maturity.

(PLACE FIG. 2 HERE)

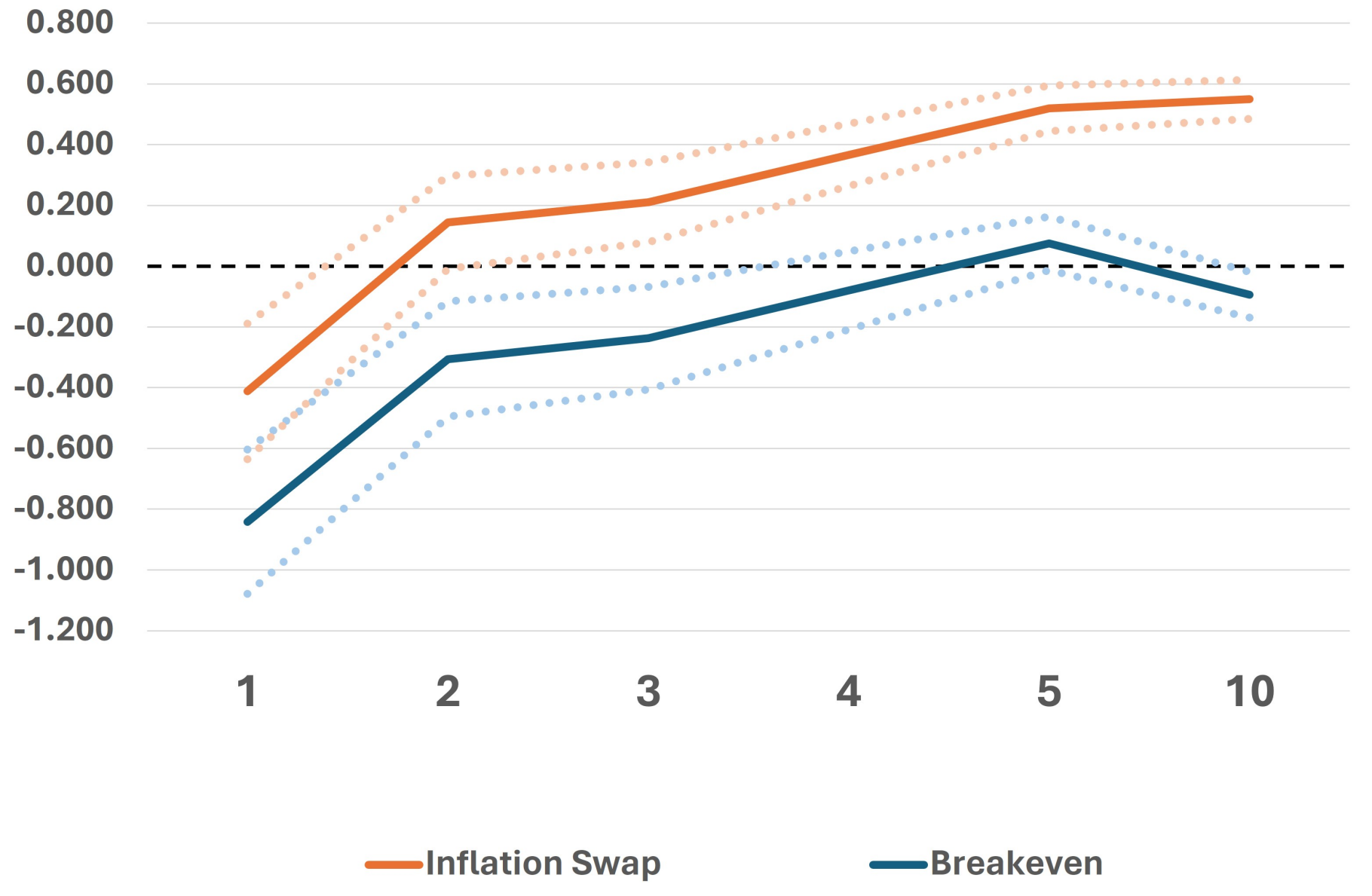
[5]Backcasting involves using current knowledge and data to estimate or revise past values of a variable or phenomenon. Nowcasting refers to the prediction or estimation of the present or very near future values of an economic or meteorological variable.

[6]Gürkaynak, Sack and Wright (2010) illustrate how to calculate the reference CPI for TIPS taking into account the indexation lag. If the maturity or issue date falls on day d_t of a month with d_n days, the reference CPI is

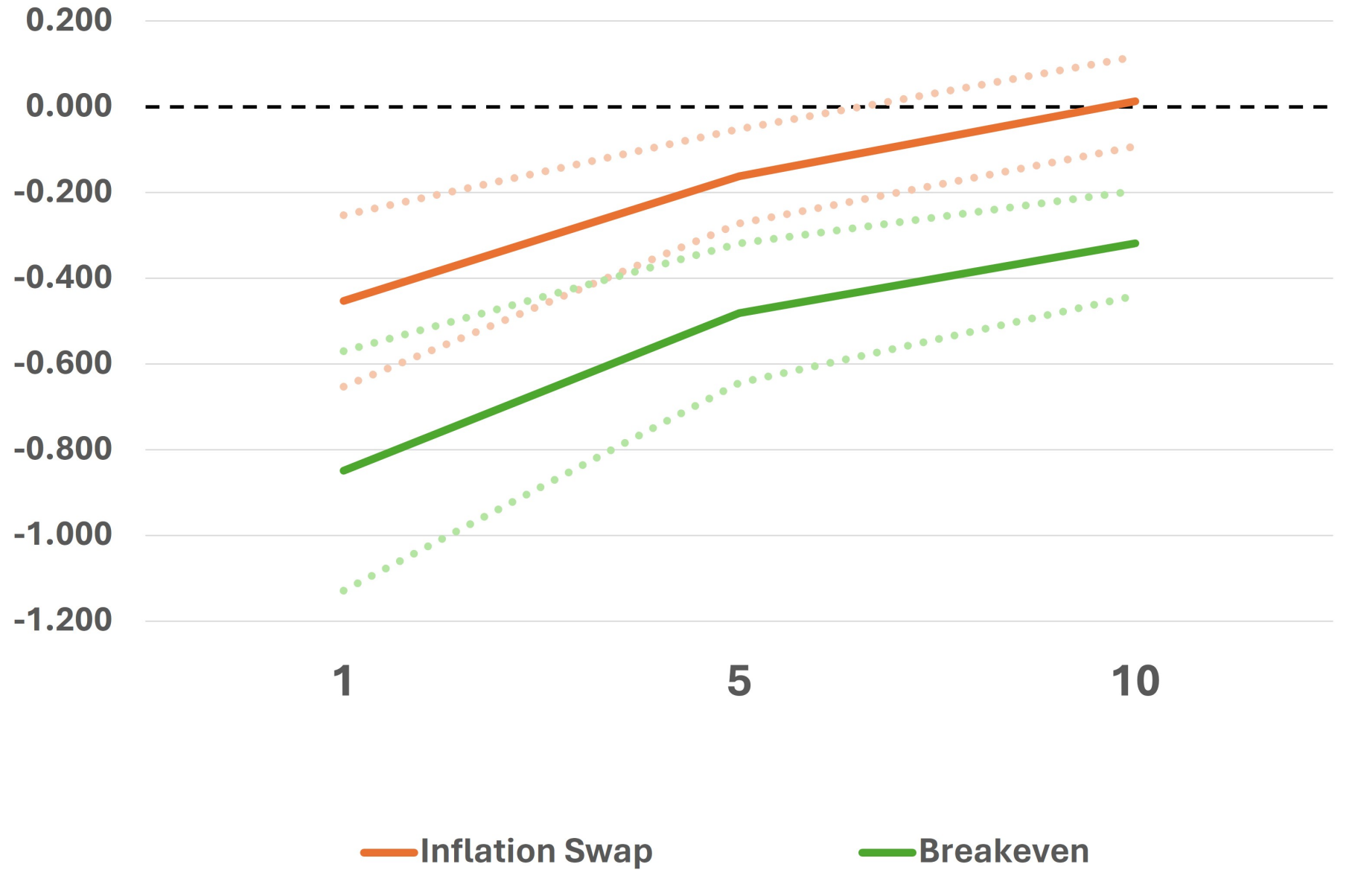
$$\text{CPI}(-2)\,\frac{d_t - 1}{d_n} + \text{CPI}(-3)\,\frac{d_n - d_t + 1}{d_n}$$

where CPI(-2) and CPI(-3) denote the non-seasonally adjusted US City Average All Items CPI for the second and third months prior to the month in which the maturity or issue date falls, respectively.

IRP1 and IRP3 (Relative to Realization)



IRP2 and IRP4 (Relative to Survey Expectation)



As will be demonstrated below, the values of IS exceed those of BE for all maturities. Moreover, as is well-known, for any non-zero volatility,

$$\text{Geometric Average} \cong \text{Arithmetic Average} - \frac{\sigma^2}{2},$$

where σ is the standard deviation of returns.

There are several inferences we can draw from Fig. 2:

1. **Rank ordering.** In terms of the rank ordering, wWe know IS > BE, and since Arithmetic > Geometric, we should expect Inflation Risk Premium$_1$ < Inflation Risk Premium$_4$.

2. **Term structure.** Across all four measures, and regardless of the benchmark, the IRP slopes upward with maturity. It is robustly negative at the one-year horizon and rises toward zero by the ten-year horizon, at which the survey-based swap IRP is essentially zero and the realization-based swap IRP even turns positive. The negative inflation risk premium is therefore strongest and most robust at the near horizon.

3. **Survey versus realization.** The two benchmarks diverge at the long end, which we attribute to the nature of realized inflation: as an ex-post measure contaminated by inflation shocks in the finite sample, it is a noisier proxy for ex-ante expectations than the direct survey forecasts, and we accordingly interpret realization-based IRP with caution.

5 Liquidity Risk

5.1 Maturity of the Market

The literature has extensively considered the relative deficiency of liquidity in TIPS market. Given this backdrop, we attempt to make the following point. Although the market for Treasury Inflation Protected Securities (TIPS) matured in early 2000s, the liquidity effect of TIPS still lingers persistently. To see the first evidence of the maturity of the TIPS market, we devised a simple test based on the following regression:

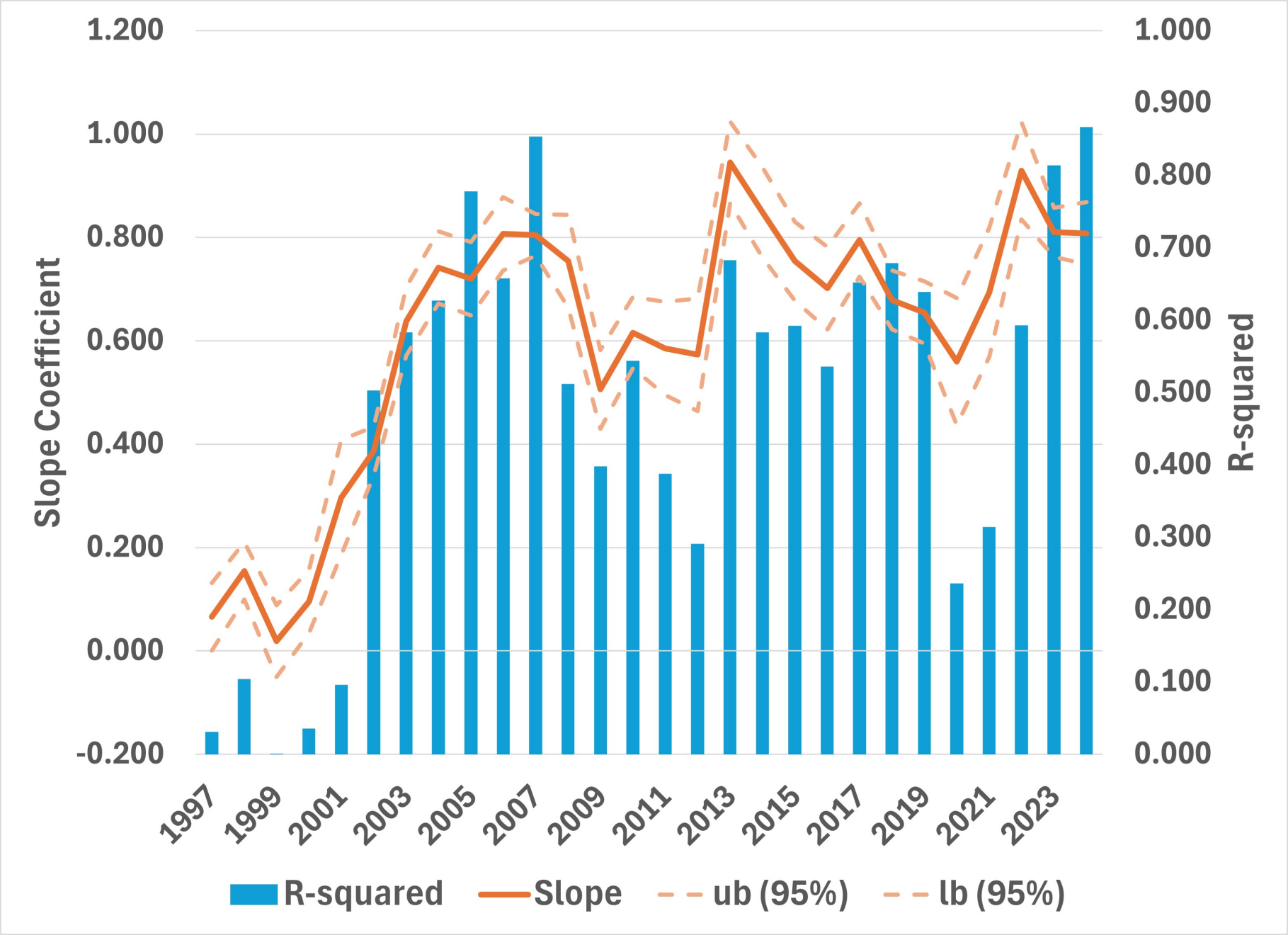
$$\Delta\,\text{GTII5}_t = \alpha + \beta \cdot \Delta\,\text{GT5}_t + \epsilon_t \tag{4}$$

$GTII5_t$ and $GT5_t$ are real and nominal Treasury bond yields. Both series are sourced from Bloomberg. The data are available at daily frequency and the sample period runs from 1997 to 5/31/2024. We run the following regression for each year from 1997 to 2024 and obtain the annual series of slope estimates $\hat{\beta}$. The intuition behind this test is that in a mature market, where the relationship between nominal yields and risk premia (including inflation and liquidity) remains stable, the regression coefficient should be slightly below unity. Detailed analytics can be found in Appendix B, which considers the analytical properties of the slope coefficient in regression (4).[7]

Figure 3 presents the result. The solid, orange line indicates the regression slope with dashed lines indicating 95% confidence intervals (left axis). The blue bars represent the explanatory power of the regression, represented by the R-squared (right axis). At the inception of the TIPS market, before 2000, the slope coefficient is close to zero, implying no systematic correlation between the nominal and real government bond markets. However, a surge in the slope and R-squared is observed around 2002, with the slope stabilizing around 0.7 after 2003. Despite fluctuations, the slope never returns to zero after 2003, suggesting the maturity of the real bond markets.

(PLACE FIG. 3 HERE)

[7]While in theory small beta estimates could reflect changes in inflation expectations rather than liquidity risk, this interpretation is unlikely given the relatively stable inflation environment in the late 1990s and early 2000s.


1.200
1.000
0.800
0.600
0.400
0.200
0.000
-0.200
Slope Coefficient
1.000
0.900
0.800
0.700
0.600
0.500
0.400
0.300
0.200
0.100
0.000
R-squared
1997
1999
2001
2003
2005
2007
2009
2011
2013
2015
2017
2019
2021
2023
R-squared
Slope
ub (95%)
lb (95%)

5.2 Lingering Effect of Illiquidity

While the evidence in the previous section indicates that the bond market, especially the inflation-indexed, has achieved some degree of maturity, it remains inconclusive whether the effects of illiquidity have fully subsided. With LRP_t representing the Liquidity Risk Premium, let us revisit the augmented Fisher equation:

$$y_t^N = y_t^R + E_t(\pi) + \text{IRP}_t - \text{LRP}_t$$

The break-even inflation rate is

$$\text{BE}_t = y_t^N - y_t^R = E_t(\pi) + \text{IRP}_t - \text{LRP}_t \tag{5}$$

This relationship suggests two potential sources of bias stemming from two types of risk premia: the inflation risk premium and the liquidity risk premium. Assuming that the inflation risk premium also exists in other inflation-related instruments with a similar magnitude, the positive liquidity risk premium would result in the break-even inflation rate being persistently below that of other inflation-related financial instruments. Another measure of inflation-indexed financial security typically compared with the break-even inflation rate is the inflation swap. However, a direct comparison with the inflation swap may not be adequate, as the two differ in the structure of coupon payments. While the inflation swap is a zero-coupon instrument, Treasury bonds issue periodic coupons. To address this issue, we construct a measure of par coupon rate using the inflation swap. Specifically, let IS_t be the zero-coupon inflation swap rate in decimal points for maturity t and c_5 the 5-yr. break-even rate also in decimal points. Then for $T = 5$, the relationship is:

$$100 = 100 \cdot c_5 * \sum_{t=1}^{5} [1/(1+IS_t)^t] + 100/(1+IS_5)^5$$

This can be solved for

$$c_5 = \frac{[1 - 1/(1+IS_5)^5]}{\sum_{t=1}^{5}[1/(1+IS_t)^t]} \tag{6}$$

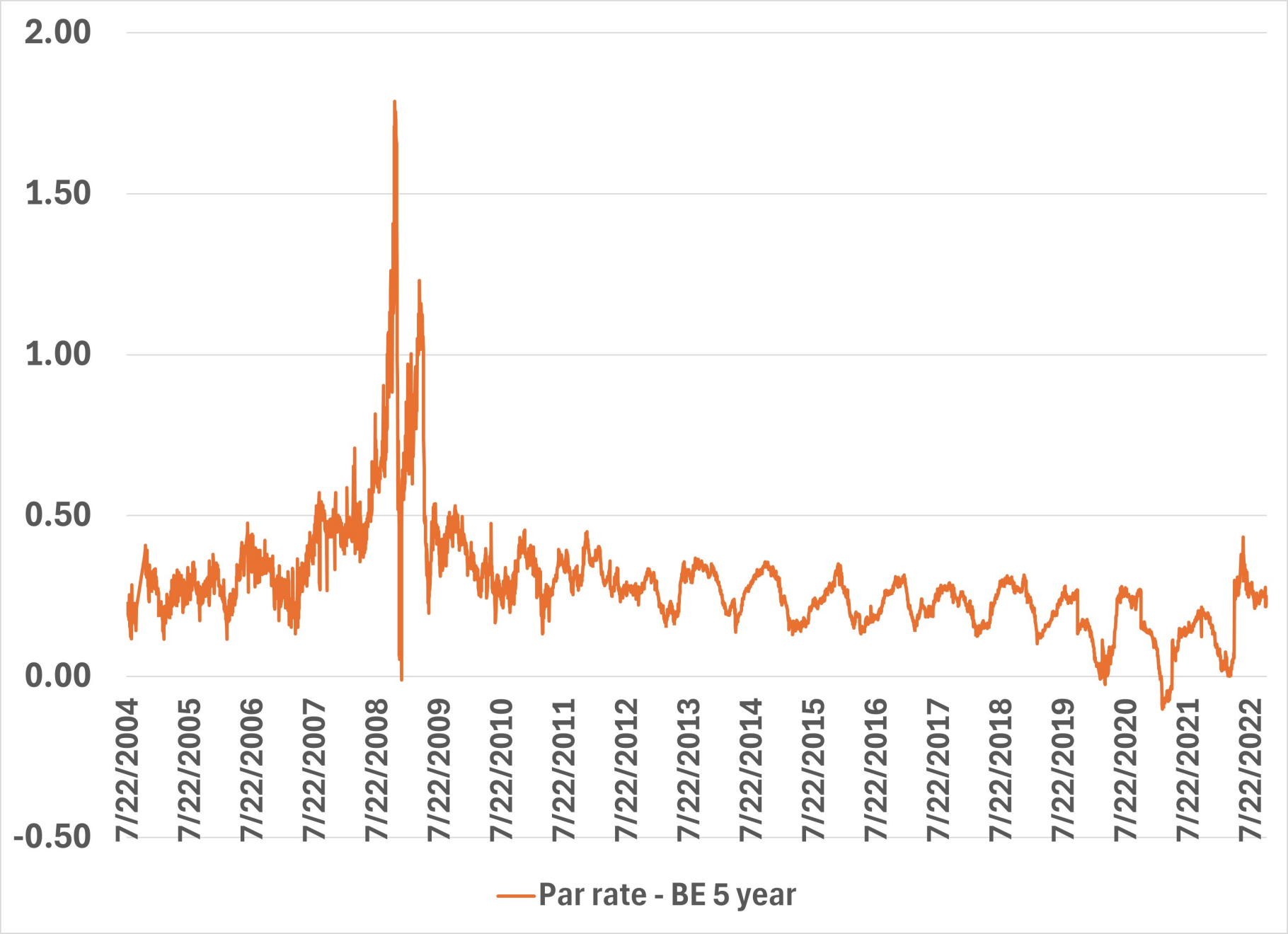
c_5 is the 5-year par rate implied by the inflation swaps. If the break-even inflation rate calculated from the nominal and real bonds systematically from the par rate, this suggests the size and the direction of the bias.

Figure 4 presents the difference between the imputed par rate and the 5-year break-even inflation rate. The sample period runs from 7/22/2004 to 10/28/2022.

The difference is persistently positive indicating a downward bias of break-even inflation rate. However, aside from the 2008 recession period, the size of the gap remains relatively stable. This mild but persistent downward bias suggests that there is a lingering effect of liquidity, even though the market has matured significantly around 2003 based on our regression evidence.

This liquidity wedge implies the empirical findings in the previous section warrant cautious interpretation: Because the inflation-swap premium at longer maturities is often close to zero or even positive, depending on the benchmark, the negative break-even premium found in BE at the long end of maturity may partly reflect liquidity rather than a genuine risk premium. At the same time, the swap market is not itself free of distortions, being subject to its own risk premia, collateral effects, and dealer balance-sheet frictions. We therefore acknowledge this cross-maturity heterogeneity and interpret the long-maturity results with appropriate caution.

(PLACE FIG. 4 HERE)


2.00
1.50
1.00
0.50
0.00
-0.50
7/22/2004
7/22/2005
7/22/2006
7/22/2007
7/22/2008
7/22/2009
7/22/2010
7/22/2011
7/22/2012
7/22/2013
7/22/2014
7/22/2015
7/22/2016
7/22/2017
7/22/2018
7/22/2019
7/22/2020
7/22/2021
7/22/2022
Par rate - BE 5 year

6 Inflation Risk Premia and Inflation Expectations Based on the CAPM

The preceding sections establish that the inflation risk premium is predominantly negative, especially at short horizons, but leave open the economic question of why investors would accept a break-even rate below expected inflation. In this regard, we believe it is crucial to examine the matter of the inflation risk premium under an explicit assumption of systematic risk. We do so under the Capital Asset Pricing Model's assumption of a single-index Market Portfolio.

Although the pricing relation resembles a static CAPM in form, its components — correlation and inflation volatility – are time-varying. Consequently, the framework is dynamic in practice: The inflation risk premium responds to evolving macroeconomic conditions and market volatility. We use the CAPM as a disciplined decomposition that explains why the sign of the premium is negative in the post-2006 regime.

6.1 Decomposition of Inflation Compensation

Assuming the classic CAPM holds,

$$E\left(r_{\text{TIPS}}\right) - r_f = \beta_{\text{TIPS}}\left[E\left(r_m\right) - r_f\right] \tag{7}$$

where

r_{TIPS} = Nominal return on TIPS security

r_f = Yield on a nominal Treasury with the same maturity as the TIPS
$= y_{\text{TREAS}}$

Let y_{TIPS} represent the real yield on TIPS. If we assume the expected change in y_{TIPS} is equal to zero, $E\left(\Delta y_{\text{TIPS}}\right) = 0$, then $E\left(r_{\text{TIPS}}\right) = y_{\text{TIPS}} + E\left(\text{Inflation}\right)$.

Since y_{TIPS} is known in advance, $\beta_{\text{TIPS}} = \beta_{\text{Inflation}}$.[8] Letting $r_f = y_{\text{TREAS}}$, we can substitute $E(r_{\text{TIPS}}) - r_f = y_{\text{TIPS}} + E(\text{Inflation}) - y_{\text{TREAS}}$ into the LHS of (7):

$$y_{\text{TIPS}} - y_{\text{TREAS}} + E(\text{Inflation}) = \frac{\text{Cov}(\text{Inflation}, r_m)}{\text{Var}(r_m)} [E(r_m) - y_{\text{TREAS}}] \tag{8}$$

If we rearrange eq. (8),

$$\begin{aligned} E(\text{Inflation}) &= y_{\text{TREAS}} - y_{\text{TIPS}} + \frac{\text{Cov}(\text{Inflation}, r_m)}{\text{Var}(r_m)} [E(r_m) - y_{\text{TREAS}}] \\ &= \text{Break-Even Inflation} + \frac{\text{Cov}(\text{Inflation}, r_m)}{\text{Var}(r_m)} [E(r_m) - y_{\text{TREAS}}] \end{aligned} \tag{9}$$

If we decompose the second expression on the RHS, we can write that as:

$$\frac{\text{Cov}(\text{Inflation}, r_m)}{\text{Var}(r_m)} [E(r_m) - y_{\text{TREAS}}] = \frac{\text{Cov}(\text{Inflation}, r_m)}{\text{Std}(r_m)} \frac{E(r_m) - y_{\text{TREAS}}}{\text{Std}(r_m)},$$

where the last expression $\frac{E(r_m) - y_{\text{TREAS}}}{\text{Std}(r_m)}$ is the Sharpe Ratio (which we will assume is a const. equal to 0.448).

We then have

[8] For this to hold, it may be sufficient to assume $\text{Cov}(y_{\text{TIPS}}, r_m)$ is equal to zero. To see this, let π_t be the day t value of the CPI-U index relevant to our measure of inflation, and write Inflation as the rate of change in π, $\frac{d\pi}{\pi}$. Now consider

$$\text{Cov}(r_{\text{TIPS}}, r_m) = \text{Cov}\left(y_{\text{TIPS}} + \frac{d\pi}{\pi}, r_m\right),$$

which reduces to $\text{Cov}\left(\frac{d\pi}{\pi}, r_m\right)$ if $\text{Cov}(y_{\text{TIPS}}, r_m) = 0$. To obtain the result $\beta_{\text{TIPS}} = \beta_{\text{Inflation}}$, we divide each expression $\text{Cov}(r_{\text{TIPS}}, r_m)$ and $\text{Cov}\left(\frac{d\pi}{\pi}, r_m\right)$ by $\text{Var}(r_m)$.

We present empirical evidence supportive of the assumption $\text{Cov}(y_{\text{TIPS}}, r_m) = 0$ in Appendix D.

$$
\begin{aligned}
E\left(\text{Inflation}\right) &= \text{Break-Even Inflation} + \frac{\text{Cov}\left(\text{Inflation},\, r_m\right)}{\text{Var}\left(r_m\right)}\left[E\left(r_m\right) - y_{\text{TREAS}}\right] \\
&= \text{Break-Even Inflation} + \frac{\text{Cov}\left(\text{Inflation},\, r_m\right)}{\text{Std}\left(r_m\right)}\,\text{Sharpe Ratio} \\
&= \text{Break-Even Inflation} + \text{Corr}\left(\text{Inflation},\, r_m\right)\,\text{Std}\left(\text{Inflation}\right)\,\text{Sharpe Ratio}
\end{aligned}
\tag{10}
$$

Eq. (10) gives rise to the computation of the Inflationn Risk Premium (IRP):

$$
\begin{aligned}
\text{Inflation Risk Premium} &\equiv \text{Break-Even Inflation} - E\left(\text{Inflation}\right) \\
&= -\,\text{Corr}(\text{ Inflation},\, r_m) \times \text{Std}(\text{Inflation}) \times \text{Sharpe Ratio}
\end{aligned}
\tag{11}
$$

The relationship provides the three components of inflation risk premia:

1. SR_t: This component represents the aggregate investor's required compensation for taking on one unit of systematic risk. Even if inflation is highly volatile (high level of risk), it would not generate a risk premium if investors were risk-neutral. Based on Ibbotson & Harrington (2021), we calibrate this unconditionally at the conventionally accepted level of 0.448.

2. $\text{Std}_t(\text{Inflation})$: This measures the magnitude of inflation uncertainty, with greater (systematic) inflation uncertainty commanding a larger risk premium. To align with the forward-looking nature of inflation expectations, we employ the implied inflation volatility from Bloomberg. The horizon of the volatility measure corresponds to the maturity of the break-even inflation rates, reflecting inflation expectations (1, 5, 10 years).

   It is noteworthy the empirical values of this variable, $\text{Std}_t(\text{Inflation})$, is relatively small. Because correlation is constrained to the unit interval, together with a bounded value for the Sharpe Ratio, this imposes an upper bound on the absolute value of the inflation risk premium.

3. $\text{Corr}_t(\text{Inflation}, r_m)$: A non-zero correlation with the market, and the resulting non-zero beta, establishes systematic exposure; the sign of the correl determines the sign of the asset's expected excess return under the CAPM. A

positive correlation implies inflation is pro-cyclical — falling during market downturns — which causes nominal bonds to gain real value precisely when investors need it most (in high marginal-utility states). Because investors view this counter-cyclical payoff as valuable insurance against recessions, they accept a break-even rate lower than expected inflation, resulting in the negative inflation risk premium we observe in the data. Given the lack of measures of the forward-looking correlation between inflation and market return, this is proxied by the moving-window, historical correlation between changes in the break-even inflation rate (or inflation swap rate) and the market return.

6.2 The Sign of the Inflation Risk Premia

The decomposition of inflation risk premia suggests that the correlation between inflation and market return is the only informative component regarding both the sign and magnitude of the risk premia. The sign of this correlation is *inversely* related to the sign of the inflation risk premia. To gauge its predictive power, we first proceed with the following regression:

$$\Delta \, \mathrm{BE}_{t,h} = \alpha_h + \beta_h \cdot \mathrm{R}_{\mathrm{SPX},t} + \epsilon_{t,h} \tag{12}$$

$\Delta \, \mathrm{BE}_{t,h}$ is the change in the break-even inflation rate (or inflation swap) with $h = 1, 5, 10$ denoting its maturity. The change in the break-even inflation rate is regressed on the market return to obtain the correlation measure, β_h. Note that the coefficients are horizon specific as we run the regression separately for each horizon. Moreover, we obtain the time series of the beta by running the regression with a 45-day moving window. The empirical results based on this time series of betas are robust to the choice of window size. In Appendix F, we recompute the key results of this paper using betas estimated at windows of 30 and 60 observations, and all results are robust.

Our theoretical object is the covariance between expected inflation and the market. Because expected inflation is not directly observable at high frequency, we follow the standard macro-finance literature in using changes in breakeven inflation rates and inflation swap rates as empirical proxies for innovations to expected inflation. A large body of work (e.g., Gürkaynak, Sack and Wright 2005; Ang, Bekaert

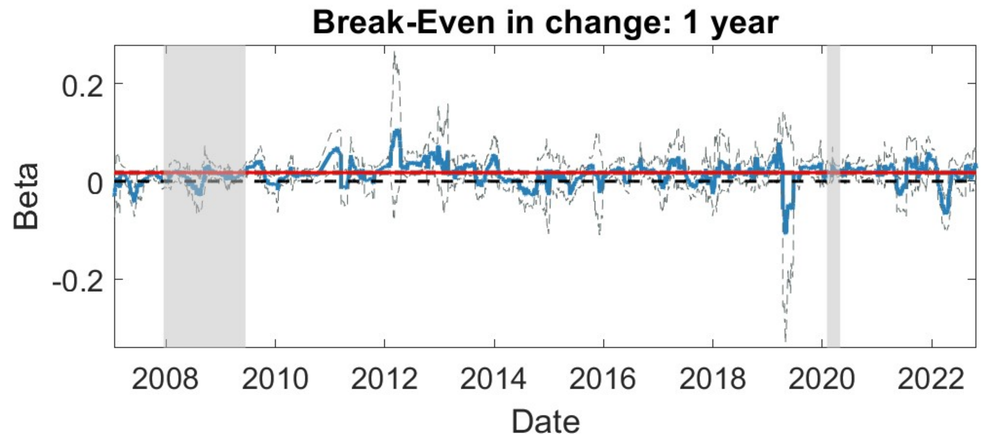
and Wei 2008; D'Amico, Kim and Wei 2018) shows that short-horizon movements in breakevens and inflation swaps are dominated by revisions to expected inflation rather than changes in inflation risk premia or liquidity components. Thus Δ BE and Δ IS provide the closest observable counterparts to the theoretical innovations in expected inflation that enter the covariance term.

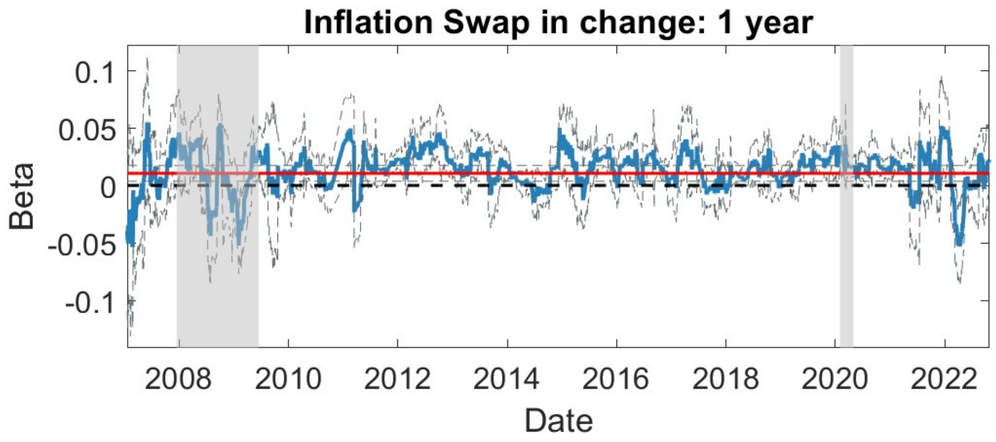
Our use of the dependent variable in eq. (12) in the form of a first-difference merits further justification. Appendix C derives the necessary conditions for this to hold. Beyond the analytics, an intuitive grounding might be useful: The independent variable is in the form of a rate of return; much as yield changes represent bonds' rates of return, we use the form of $\Delta\,\mathrm{BE}_{t,h}$ changes instead of levels because changes more closely align with the concept of regressing individual security returns on market returns in the CAPM regression.

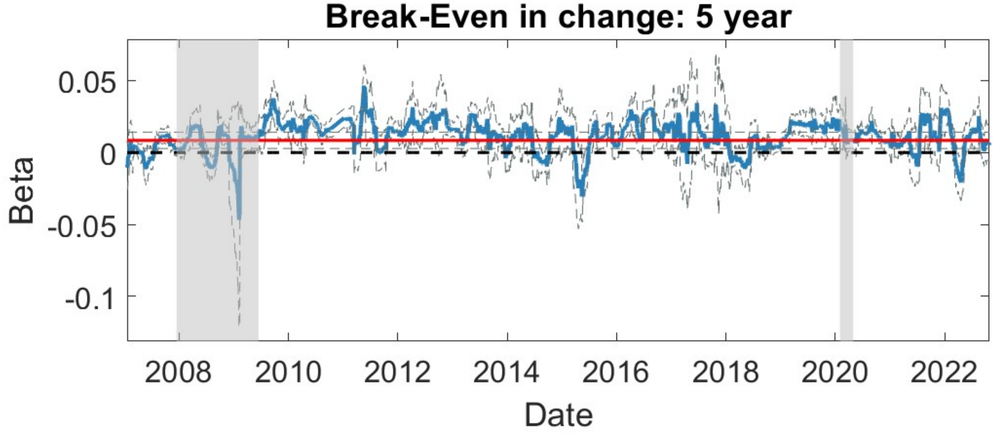
Figure 5 shows the time series of β_h obtained from the CAPM rolling-window regression. The left and right panels present results using the changes in the breakeven inflation rates and inflation swap rates, respectively. Each row corresponds to maturities of 1, 5, and 10 years. The red solid line in each subplot indicates the full sample average of β_h, while the dashed lines represent the 95% confidence bands constructed from the Newey-West standard errors.

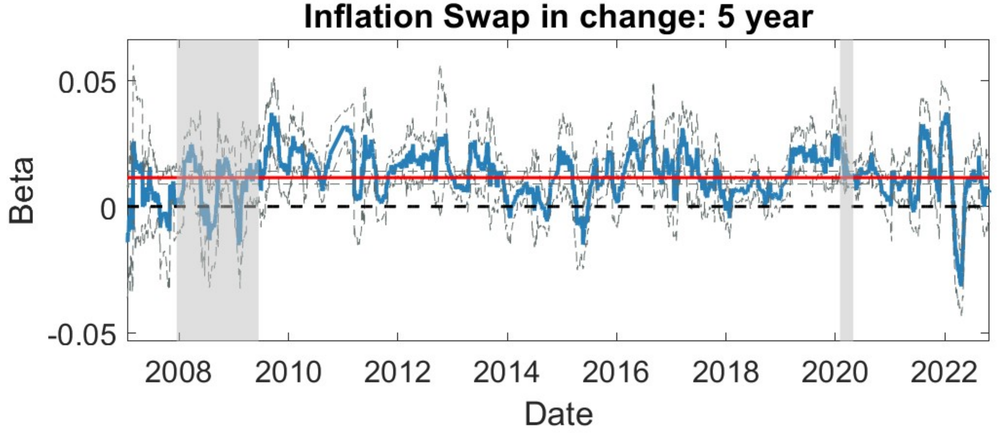
The time series of β_h presented in Figure 4 is persistently positive. Given the inverse relationship with the sign of the inflation risk premia, this time variation suggests the presence and pervasiveness of a negative inflation risk premia consistent with our earlier evidence.

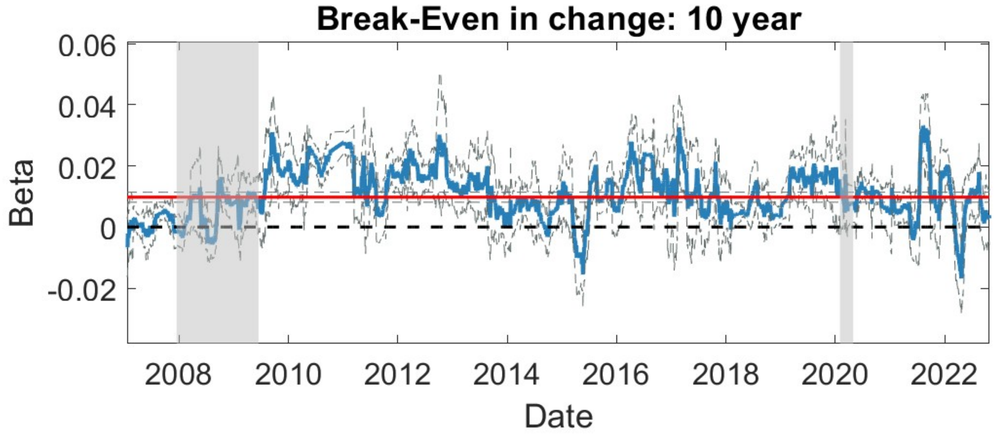
(PLACE FIG. 5 HERE)

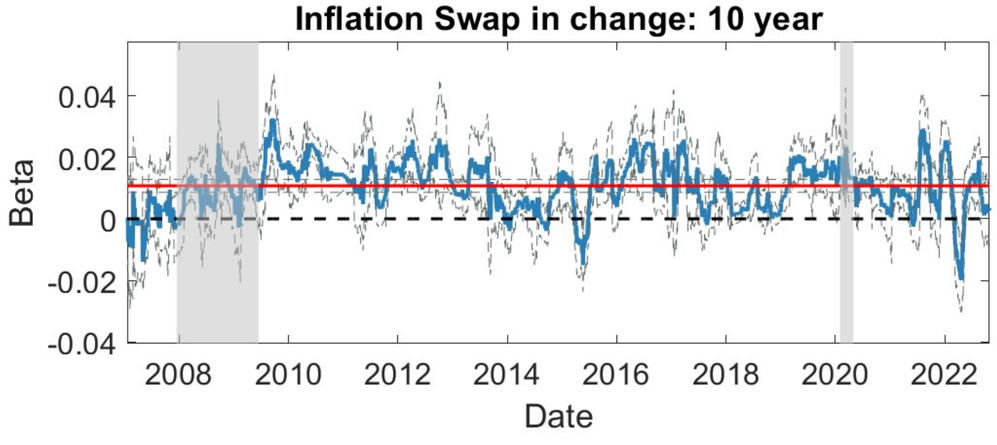

Break-Even in change: 1 year
Beta
0.2
0
-0.2
2008
2010
2012
2014
2016
2018
2020
2022
Date


Inflation Swap in change: 1 year
Beta
0.1
0.05
0
-0.05
-0.1
2008
2010
2012
2014
2016
2018
2020
2022
Date


Break-Even in change: 5 year
Beta
0.05
0
-0.05
-0.1
2008
2010
2012
2014
2016
2018
2020
2022
Date


Inflation Swap in change: 5 year
Beta
0.05
0
-0.05
2008
2010
2012
2014
2016
2018
2020
2022
Date


Break-Even in change: 10 year
Beta
0.06
0.04
0.02
0
-0.02
2008
2010
2012
2014
2016
2018
2020
2022
Date


Inflation Swap in change: 10 year
Beta
0.04
0.02
0
-0.02
-0.04
2008
2010
2012
2014
2016
2018
2020
2022
Date

The CAPM regression suggests that the time series of β_h should be informative of the sign of the inflation risk premia. To check this, we compare the sign of β_h with the sign of the inflation risk premia measured from survey expectations and realizations ($\text{IRP}_{1,h}$ through $\text{IRP}_{4,h}$).[9]

Table 2 presents the results of the sign concurrence analysis. Each column represents the data used to construct the time series of β_h. Panels A and B indicate which type of inflation risk premia is being compared. Each figure in the table represents the ratio of sign concurrence. For example, the upper left element of 0.755 indicates that the sign concurrence ratio between β_h, obtained from the regression of changes in the 1-year break-even inflation rate on market returns, and the inflation risk premia, measured as the difference between the 1-year break-even inflation rate and the 1-year expectation from the SPF ($\text{IRP}_{2,1}$), is 75.5%. The rows labeled "Unconditional" and "Conditional" indicate whether the sign concurrence was measured without regard to statistical significance or conditional on statistically significant betas at the 95% level, respectively.

The result in Panel A suggests that the betas obtained from the CAPM regression are very informative of the sign of the risk premia.The result especially holds for the IRP from the SPF ($\text{IRP}_{2,h}$ and $\text{IRP}_{4,h}$). The ratio of sign concurrence declines over maturity but remains large. Conditional only on the statistically significant betas, the ratio improves almost to unity for the 1-year horizon.

In contrast, Panel B suggests that the betas from the market measures do not align as much with the IRP from the realization. In general, the ratios are much smaller than the ratios in Panel A. suggesting that the betas obtained from the changes in BE and IS are more consistent with the survey expectation rather than the actual realization.

[9]To align the different frequencies of the daily betas with the quarterly series of survey-based inflation risk premia, we use the SPF deadline date as the reference date.

Table 2: Sign Concurrence between $\beta_{\text{SPX},\,\Delta\text{BE},t}$ and IRP from the SPF

	break-even			Inflation Swap		
	1 year	5 year	10 year	1 year	5 year	10 year
A. IRP from the SPF						
Unconditional	0.755	0.849	0.849	0.887	0.660	0.434
Conditional	0.917	0.947	0.872	1.000	0.725	0.500
B. IRP from Realizations						
Unconditional	0.691	0.408	0.254	0.611	0.383	0.089
Conditional	0.894	0.384	0.271	0.639	0.372	0.063

Notes:

1. This table reports the result on the sign concurrence between the correlation between changes in BE (IS) and market return, and the inflation risk premia ($\text{IRP}_{1,h}$ through $\text{IRP}_{4,h}$). The correlation measure is obtained using a 45-day moving-window regression of changes in BE (IS) on market returns (S&P 500). The resulting daily series of betas are then matched with the time series of IRP.
2. Panel A reports the sign concurrence with the IRP from the survey ($\text{IRP}_{2,h}$ and $\text{IRP}_{4,h}$).
3. Panel B reports the sign concurrence with the IRP from the realization ($\text{IRP}_{1,h}$ and $\text{IRP}_{3,h}$).
4. The frequencies for Panel A and Panel B are quarterly and daily, respectively.
5. Unconditional refers to the ratio where betas are used regardless of statistical significance whereas Conditional refers to the ratio when only statistically significant betas were used. Statistical significance was assessed using Newey-West standard errors at 95% level.
6. The sample period runs from 2009Q2 – 2022Q3 for Panel A and 2009.05.12 – 2022.10.26 for Panel B

Additionally, it raises the question of comparing the betas obtained from forward-looking market measures such as BE and IS with those derived from realized inflation regarding sign concurrence. To investigate this, we conduct a similar sign concurrence exercise using betas obtained from realized inflation rates:

$$\pi_{t,h} = \alpha_h + \beta_{\text{SPX},\pi,h} \cdot R_{\text{SPX},t} + \epsilon_{t,h} \tag{13}$$

where $\pi_{t,h}$ is realized h-year inflation. The regression is run with a 45-month moving window to obtain the time series of horizon-specific betas. Table 3 presents the results of the sign concurrence between $\beta_{\text{SPX},\pi,h}$ and the inflation risk premia. One notable change is observed in terms of the performance of sign concurrence with the IRP based on survey expectations (Panel A. IRP from the SPF). While the betas obtained from changes in BE and IS had a sign concurrence close to unity, the sign concurrence of betas from realized inflation ranges only from 0.4 to 0.7, even when considering only statistically significant betas at the 95% level. This confirms that the betas obtained from market measures are particularly useful for predicting the sign of the inflation risk premia measured from survey expectations.[10]

[10]In running regressions of estimated betas from changes in BE and IS on survey-based IRP we obtained statistically weak results.

Table 3: Sign Concurrence between $\beta_{\text{SPX},\pi,t}$ and IRP from the SPF

	break-even			Inflation Swap		
	1 year	5 year	10 year	1 year	5 year	10 year
A. IRP from the SPF						
Unconditional	0.660	0.566	0.547	0.698	0.547	0.547
Conditional	0.719	0.618	0.500	0.688	0.559	0.450
B. IRP from the Realization						
Unconditional	0.731	0.506	0.175	0.688	0.500	0.231
Conditional	0.771	0.520	0.000	0.677	0.471	0.016

Notes:

1. This table reports the result on the sign concurrence between the correlation between monthly CPI inflation and market return and the inflation risk premia ($\text{IRP}_{1,h} - \text{IRP}_{4,h}$). The correlation measure is obtained using a 45-month moving-window regression of inflation on market returns (S&P 500). The resulting monthly series of betas are then matched with the time series of IRP.
2. Panel A reports the sign concurrence with the IRP from the survey ($\text{IRP}_{2,h}$ and $\text{IRP}_{4,h}$).
3. Panel B reports the sign concurrence with the IRP from the realization ($\text{IRP}_{1,h}$ and $\text{IRP}_{3,h}$).
4. The frequencies for Panel A and Panel B are quarterly and monthly, respectively. Unconditional refers to the ratio where betas are used regardless of statistical significance whereas Conditional refers to the ratio where only statistically significant betas were used. Statistical significance was assessed using Newey-West standard errors at 95% level.
5. The sample period runs from 2009Q2 - 2022Q3 for Panel A and 2009M5 - 2022M8 for Panel B.

6.3 The Magnitude of the Inflation Risk Premia

Table 4 presents a comparative analysis of the inflation risk premia (IRP) measured via three distinct methods across different maturities (1 year, 5 years, and 10 years). Panel A. Realization refers to the IRP measured by taking the difference between the BE (IS) and the realized inflation using reference CPI ($\text{IRP}_{1,h}, \text{IRP}_{3,h}$). Panel B. Survey presents descriptive statistics related to the IRP measured by taking the difference between the BE (IS) and the inflation expectations from the SPF ($\text{IRP}_{2,h}, \text{IRP}_{4,h}$). Lastly, Panel C. Survey presents the results of the IRP measured from the CAPM model.

The findings reveal a notable divergence of the CAPM IRP from the other measures in the term structure across maturities. As aforementioned, there is an upward-sloping term structure for IRP directly measured from the realized inflation and survey expectations. This is indicated by the fact the IRP becomes less negative (or even positive for IS) as the maturity increases. Specifically, for the IRP from the realization, the mean IRP for the 1-year maturity is -1.183 for BE and -0.782 for IS, which gradually increases to -0.198 and 0.411 for the 10-year maturity, respectively. A similar pattern is observed for the IRP compared with the survey. IRP becomes less negative as maturity increases. This upward slope suggests that the perceived inflation risk decreases with longer maturities, potentially reflecting larger stability in inflation and inflation expectations in longer horizons.

In contrast, the IRP from the CAPM exhibits a downward-sloping term structure. The mean IRP for the 1-year maturity is -0.135 for BE and -0.171 for IS, which grows larger, reaching -0.634 and -0.620 for the 10-year maturity, respectively. This inverse relationship may be due to higher values of implied volatility (ZC volatility) of inflation for longer maturities.

Additionally, the volatility (Std.) of the IRP also shows opposite term structures: it decreases with maturity for the realization and survey but increases for the CAPM. This contrast in findings highlights the differences in how each method quantifies inflation risk.

Table 4: The Magnitude of the Inflation Risk Premia (%)

	break-even			Inflation Swap		
	1 year	5 year	10 year	1 year	5 year	10 year
A. Realization						
Mean.	−1.183	−0.549	−0.198	−0.782	−0.134	0.411
Std.	1.653	1.124	0.515	1.644	1.050	0.466
B. Survey						
Mean.	−0.806	−0.425	−0.295	−0.404	−0.158	−0.004
Std.	0.859	0.305	0.289	0.450	0.296	0.283
C. CAPM						
Mean.	−0.135	−0.388	−0.634	−0.171	−0.418	−0.620
Std.	0.173	0.319	0.440	0.201	0.273	0.433

D. CAPM	Realized Inflation		
	1 year	5 year	10 year
Mean.	−0.117	−0.101	−0.043
Std.	0.204	0.484	0.512

Notes:

1. This table presents the descriptive statistics of different time series of inflation risk premia (IRP).
2. Panel A. Realization refers to the IRP computed by taking the difference between BE (or IS) and realized inflation using CPI-U data ($\text{IRP}_{1,h}$ and $\text{IRP}_{3,h}$).
3. Panel B. Survey refers to the IRP, computed by taking the difference between BE (or IS) and the survey inflation expectation from the SPF data ($\text{IRP}_{2,h}$ and $\text{IRP}_{4,h}$).
4. Panel C. CAPM refers to the IRP from the CAPM model that involves the betas obtained by using the changes in BE (and IS).
5. Panel D. CAPM refers to the IRP obtained from the CAPM model that involves the betas obtained by using realized inflation.
6. Each column specifies the data used and the maturity. The sample period runs from 2009Q2 to 2022Q3, with dates aligned based on the SPF survey deadline.

To make the sources of the downward-sloping term structure of the CAPM-based inflation risk premium transparent, we provide an exact decomposition by maturity. Since the magnitude of the CAPM premium is the product of three terms,

$$-\text{IRP}(h) = \text{Corr}(\text{Inflation, r}_\text{m}) \times \text{Std}(\text{Inflation}) \times \text{SR},$$

taking logs renders it additive, and differencing 5- and 10-year horizons relative to the one-year horizon, with the Sharpe ratio held constant across maturities, yields

$$\Delta \log(-\text{IRP}) = \Delta \log \text{Corr}(\text{Inflation, r}_\text{m}) + \Delta \log \text{Std}(\text{Inflation}).$$

$\Delta \log(-\text{IRP})$ refers to the difference between 5- or 10-year IRP and one-year IRP. Each component's contribution to the change in the premium across maturities is therefore exact, with no interaction term, and the two shares sum to one hundred percent. The results are reported in Table 5.

Table 5 — Decomposition of the CAPM-based Inflation Risk Premium

	break-even		inflation swap	
	5yr	10yr	5yr	10yr
$\Delta \log(-\text{IRP})$ (vs. 1yr)	1.100	1.623	0.972	1.349
$\Delta \log \text{Corr}(\text{Inflation}, r_m)$	0.471	0.607	0.342	0.333
$\Delta \log \text{Std}(\text{Inflation})$	0.629	1.016	0.629	1.016
Share: Corr(Inflation, r_m)	42.8%	37.4%	35.2%	24.7%
Share: Std(Inflation)	57.2%	62.6%	64.8%	75.3%

Notes:

1. $-\text{IRP} = \text{Corr}(\text{Inflation, r}_\text{m})\,\text{Std}(\text{Inflation})\,\text{SR}$, with $\text{SR} = 0.448$ constant.
2. Entries are the 1yr $\rightarrow h$ log changes and the implied shares, which sum to 100%. The sum of second and third rows should match the numbers in the first row.
3. Corr(Inflation, r_m) is $\text{Corr}(\Delta\text{BE}, r_m)$ for break-even and $\text{Corr}(\Delta\text{IS}, r_m)$ for the inflation swap.

The decomposition shows that the steepening of the term structure is driven primarily by the rising implied (zero-coupon) volatility of inflation, which accounts for 57 to 63 percent of the change for break-even and 65 to 75 percent for the inflation swap. The correlation between inflation and the market return accounts for the remainder, and the constant Sharpe ratio contributes nothing by construction. The downward-sloping term structure of the CAPM-based premium is thus not an artifact of the Sharpe-ratio calibration, but reflects the term structure of priced inflation risk, inherited from the upward-sloping volatility curve of inflation.

This also accounts for the opposite-sloping term structures of the CAPM-based and survey-based premia. The two rest on different views of how inflation uncertainty evolves with horizon. As Figure 1 shows, the market's implied inflation volatility rises with maturity while the dispersion of survey forecasts falls: the marginal investor perceives more inflation risk the further out one looks, whereas professional forecasters increasingly agree on a stable long-run trend anchored near 2 percent. Because the CAPM premium is compensation for bearing this systematic risk, it grows more negative as priced risk accumulates with horizon. The survey-based premium narrows toward zero as the consensus settles. The divergence is thus not a contradiction but a reflection of two distinct objects: the price of inflation risk to the marginal investor versus the representative forecaster's central expectation. Reassuringly, the two approaches agree where it matters most for our central result: at the near horizon, both point to a clearly negative premium.

Beyond the level and term structure, we can ask whether the CAPM mechanism itself — the beta — identifies the premium. The sign concurrence test provides suggestive mechanism of negative inflation risk premia by showing that the estimated betas mostly align with the survey-based inflation risk premia, especially at the near-horizon using the inflation swap betas. In addition, we also show that the beta is capable of providing predictability in terms of magnitude. To this end we run the following regression:

$$\mathrm{IRP}_{4,h,t} = a + b\,\beta_{\Delta IS,h,t} + e_{t,h}$$

The results are reported in Table 6.[11]

[11] Appendix E replicates these tests for the Breakeven Rates of inflation.

Table 6 — Predictability Regression: Inflation Swap

Horizon	n	conc.	Slope b	t	p	R^2
1yr	53	0.887	−0.158	−2.37	0.018	0.123
5yr	53	0.660	−0.010	−0.29	0.774	0.001
10yr	53	0.434	−0.044	−1.18	0.239	0.024

Notes:

1. The estimated regression result from $\text{IRP}_{4,h} = a + b\,\beta_{\Delta\text{IS},h}$.
2. The regressor is standardized by its own standard deviation. So b is the pp change in the IRP per one standard deviation of beta.
3. "conc." is the sign-concurrence ratio, which uses the same observations as the regression.
4. t and p are t-statistics and p-values computed using Newey-West standard errors.

We report the results based on inflation-swap betas here because the swap beta is the cleaner measure, less exposed to TIPS liquidity. The break-even counterpart is presented and discussed in the Appendix E.

Consistent with the sign-concurrence evidence, the predictive relationship is concentrated at the near horizon. At one year, the beta is a statistically significant, correctly-signed predictor of the survey premium: a one-standard-deviation increase in beta lowers the premium by about 0.16 percentage points. At five and ten years the slope keeps the theory-predicted negative sign but is small and statistically insignificant. The magnitude evidence thus mirrors the sign-concurrence ratios, which fall from 0.89 at one year to 0.43 at ten: the betas carry both directional and quantitative information about the premium, but most clearly at the short horizon.

6.4 Risk-Adjusted Inflation Expectation

In this subsection, we compare the following series of inflation expectations.

1. Inflation expectation from the SPF.

2. Risk-adjusted inflation expectation from the CAPM using BE and IS

$$\begin{aligned} E(\text{inflation}) &= y_{\text{TREAS}} - y_{\text{TIPS}} + \underbrace{\text{Corr}(\Delta\,\text{BE},\, r_m) \times \text{Std}(\text{Inflation}) \times \text{SR}}_{\text{–IRP}} \\ E(\text{inflation}) &= y_{\text{TREAS}} - y_{\text{TIPS}} + \underbrace{\text{Corr}(\Delta\,\text{IS},\, r_m) \times \text{Std}(\text{Inflation}) \times \text{SR}}_{\text{–IRP}} \end{aligned} \tag{14}$$

3. Risk-adjusted inflation expectation from the CAPM using realizations

$$E(\text{inflation}) = \begin{pmatrix} \text{Realized} \\ \text{Inflation} \end{pmatrix} + \underbrace{\text{Corr}(\text{Inflation},\, r_m) \times \text{Std}(\text{Inflation}) \times \text{SR}}_{\text{–IRP}}. \tag{15}$$

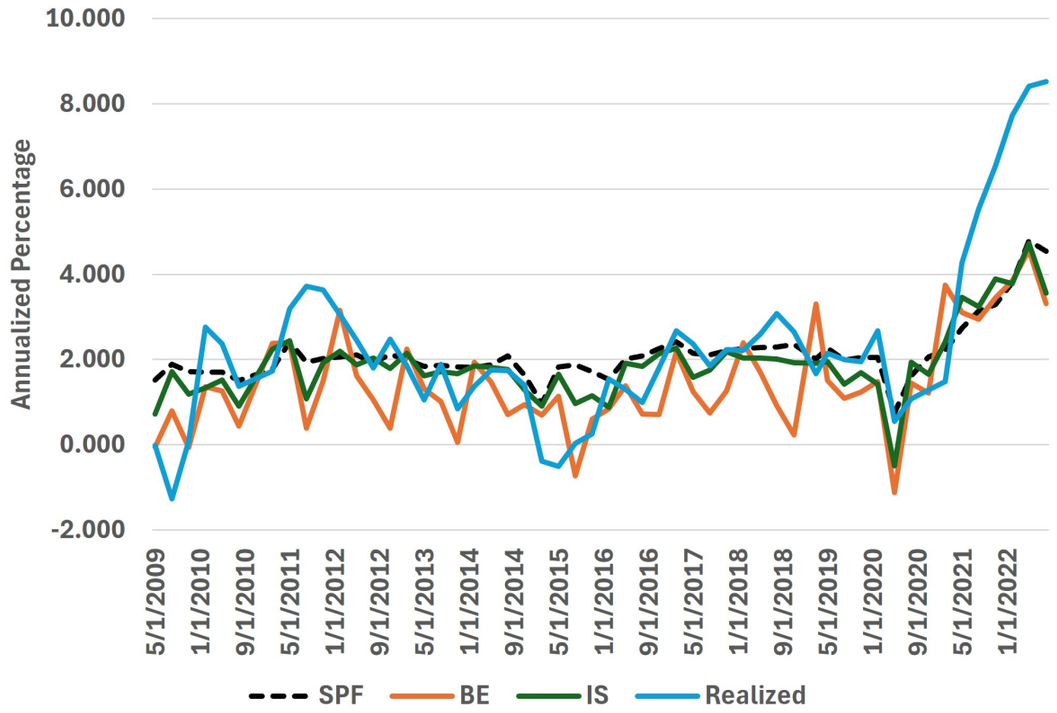
The maturities we consider is 1, 5 and 10 years.

Figure 6 presents the time series of inflation expectations from different sources, with each subplot representing maturities of 1, 5 and 10 years. The dashed black lines in each subplot indicate the inflation expectations from the SPF, while the solid lines in different colors represent risk-adjusted inflation expectations from the CAPM model.

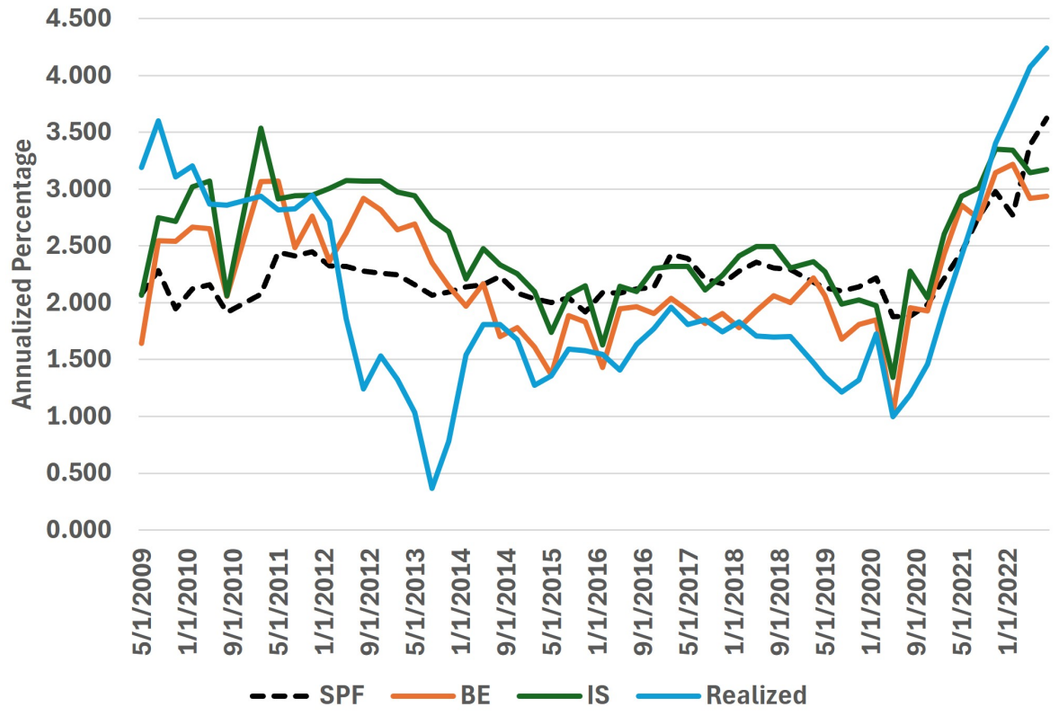
The figure reveals that the risk-adjusted inflation expectations are closely correlated with the survey expectations, especially at the short horizon of 1 year. Market-based, risk-adjusted inflation expectations using BE (solid orange) and IS (solid green) are more volatile than and tend to fluctuate around survey expectations. These measures also capture the uptick in survey expectations during the post-COVID inflation surge in 2021. The risk-adjusted expectation using realization (blue solid line) tends to be the most volatile. Following the recent rise in inflation, it shows the largest surge which is more excessive than the survey expectation.

(PLACE FIG. 6 HERE)

Inflation Expectation - 1 year ahead



Inflation Expectation - 5 year ahead



Inflation Expectation - 10 year ahead

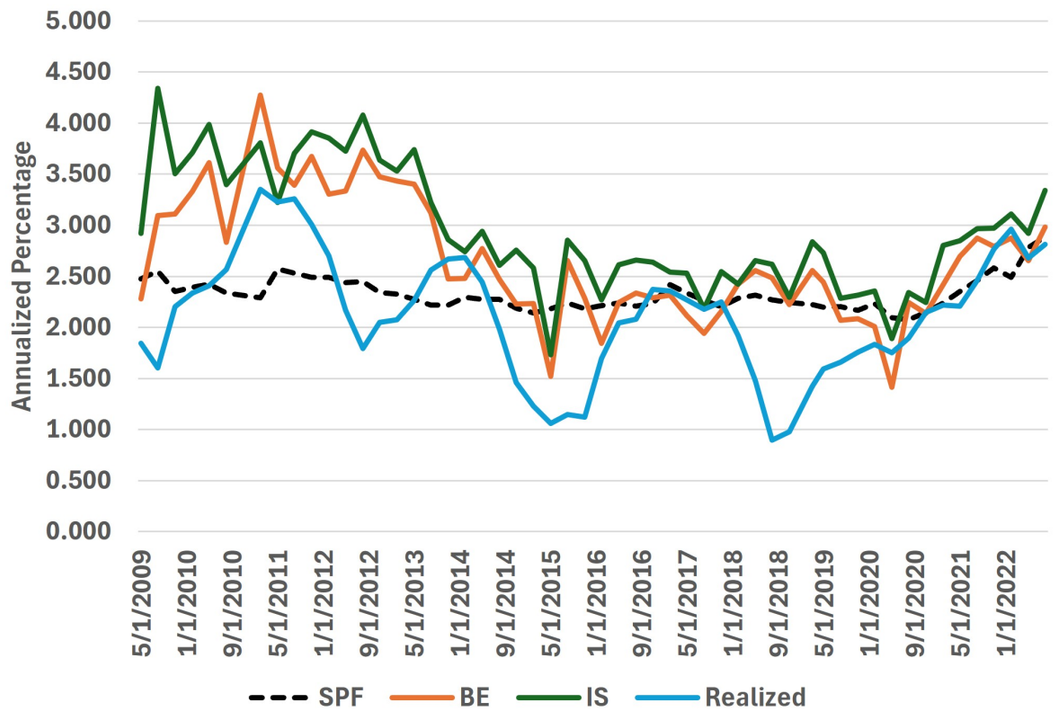


Table 7 presents the descriptive statistics of the time series presented in Figure 5. For each maturity, the table compares the mean, standard deviation, maximum, median, and minimum values of inflation expectations derived from the SPF, BE, IS, and realized inflation. The mean inflation expectations generally increase with maturity but stay within the range of 2-3 percent. The standard deviation of the survey expectation (SPF) decreases as the horizon increases. This resonates with the term structure of forecast dispersion where disagreement among forecasters tends to decrease as the horizon increases. This reflects the fact that the long-run inflation expectation in the survey is more anchored compared to the short-run inflation expectation. The table also reiterates the previous finding that the realized measure exhibits the highest volatility across all maturities, reflecting significant fluctuations in actual inflation.

Table 7 — Descriptive Statistics of Inflation Expectations (%)

	1 Year				5 Year				10 Year			
	SPF	BE	IS	Realized	SPF	BE	IS	Realized	SPF	BE	IS	Realized
Mean.	2.132	1.462	1.899	2.260	2.260	2.223	2.520	2.037	2.327	2.665	2.942	2.111
Std.	0.698	1.174	0.857	2.010	0.329	0.509	0.485	0.877	0.161	0.602	0.600	0.592
Max.	4.790	4.569	4.723	8.524	3.623	3.218	3.535	4.242	2.887	4.276	4.342	3.353
Median.	2.027	1.260	1.844	1.857	2.168	2.060	2.412	1.746	2.277	2.487	2.836	2.165
Min.	0.741	−1.124	−0.495	−1.273	1.876	1.020	1.344	0.367	2.073	1.413	1.733	0.896

Notes:

1. This table presents the descriptive statistics of different time series of inflation expectations. SPF denotes the inflation expectation from the Survey of Professional Forecasts. BE and IS represent the inflation expectations obtained from the CAPM model using break-even and Inflation Swap rates, respectively.

2. Realized refers to the inflation expectation from the CAPM model based on realized inflation using CPI-U data.

3. The sample period runs from 2009Q2 to 2022Q3, with dates aligned based on the SPF survey deadline.

Lastly, Table 8 reports the correlations among the different inflation-expectation series, combining three correlation matrices, one for each maturity. The SPF is strongly and positively related to the other measures, especially at the one-year horizon, where its correlation with the market-based, risk-adjusted expectations is highest — 0.904 for the inflation swap. These correlations weaken as maturity lengthens but remain positive throughout. The inflation swap's especially tight alignment with the survey at one year is consistent with our earlier results: the one-year horizon is also where the sign-concurrence ratio is highest and where the predictability regression is statistically significant. More broadly, the generally high correlations between the CAPM-based risk-adjusted expectations and the survey expectation lend support to our approach to quantifying the magnitude of the inflation risk premium.

While the correlation between the CAPM-implied long-run expectation and the survey's remains high, its weakening at longer horizons merits some discussion. This may reflect the compensation the marginal investor requires for bearing long-horizon inflation risk. As Figure 6 suggests, at the five- and ten-year horizons, the market-based, risk-adjusted expectation sits somewhat above the representative forecaster's consensus, which is firmly anchored near 2 percent. This gap mirrors the CAPM premium becoming more negative with maturity. Even so, the divergence is modest. The two series remain close and move together, a difference in degree rather than in kind between the marginal investor's and the representative forecaster's long-run views.

In sum, the market-based, risk-adjusted inflation expectations from the CAPM track the inflation expectation from the survey data quite remarkably. This is especially true for the inflation expectation in the nearest maturity.

Table 8 — Correlation Matrix of Inflation Expectations

	1 year				5 year				10 year			
	SPF	BE	IS	Realized	SPF	BE	IS	Realized	SPF	BE	IS	Realized
SPF	1.000				1.000				1.000			
BE	0.720	1.000			0.585	1.000			0.528	1.000		
IS	0.904	0.849	1.000		0.575	0.950	1.000		0.570	0.924	1.000	
Realized	0.888	0.700	0.802	1.000	0.651	0.588	0.562	1.000	0.537	0.556	0.468	1.000

Notes:

1. This table presents the correlation between different time series of inflation expectations. SPF denotes the inflation expectation from the Survey of Professional Forecasts. BE and IS represent the inflation expectations obtained from the CAPM model using break-even and Inflation Swap rates, respectively.

2. Realized refers to the inflation expectation from the CAPM model based on realized inflation using CPI-U data.

3. The sample period runs from 2009Q2 to 2022Q3, with dates aligned based on the SPF survey deadline.

6.5 Summary and Discussions

Based on our systematic-risk, CAPM analysis, we believe several conclusions apply:

1. Simlar to the unconditional estimates presented in Section 5, the systematic-risk analysis reinforces the notion of a negative IRP, most clearly at the near horizon: The break-even and inflation-swap rates of inflation are lower than the corresponding expected rates.

2. These negative values persist at all maturities save for the very longest, ten years, at which positive numbers are encountered for the realization-based swap premium.

   For our part, we take an especially careful analysis of the CAPM's implications for the inflation risk premium, and the consistency of those results with those obtained from the Survey of Professional Forecasters.

   We recognize survey-based expectations, such as the SPF, provide a direct measure of expected inflation and therefore a natural benchmark. At the same time, market-based measures such as breakeven inflation and inflation-swap rates incorporate real-time information, and thus reflect the marginal investor's valuation of inflation risk. Because both approaches have strengths and limitations, we view them as complementary rather than hierarchical. Importantly, Tables 2 – 5 show that SPF forecasts, BE, IS and realized inflation all move closely together across most horizons. This concordance suggests that the CAPM-implied term structure is not contradicting the direct measures, but rather is consistent with them when interpreted jointly.

   To address this term-structure issue, additional empirical analysis, decomposing the source of CAPM-based IRP, demonstrated that the rising ZC volatility of inflation is the primary driver. Further, the clearest explanation rests on the distinction between the survey's representative forecaster and the market's marginal investor, which is also why the two measures are complementary rather than competing. The SPF reports the representative forecaster's expectation, which is anchored near 2% at long horizons. Break-even and swap rates are prices set by the marginal investor, who must be compensated for

bearing inflation risk.

The flip side of the CAPM premium growing more negative with maturity is visible in the expectations themselves: The CAPM-implied long-run inflation expectation lies above the survey's (see the 5- and 10-year series in Figure 6. The marginal investor's risk-adjusted long-run expectation exceeds the anchored consensus, consistent with the marginal investor pricing more long-run inflation risk and possibly holding a higher long-run mean than the anchored forecasters.

3. One important qualitative result of our analysis can be found in eqs. (14) and eq. (C1): There is a strong upper bound to the inflation risk premium, found in the product of the terms Corr(Δ BE, r_m) $\times$ Std(Inflation) $\times$ SR.

   As demonstrated in Fig. 1 above, the empirical regularity from the Bloomberg system shows the traded values of Std(Inflation) are for the most part lower than 3%. With that volatility input, we can compute the upper bound Corr(Δ BE, r_m) $\times$ Std(Inflation) $\times$ SR, while recognizing the absolute value of the correlation is bounded by +1, and we clearly have an accurate sense of what the Sharpe Ratio is. While that correlation of +1 is significantly higher than empirically observed correlations, even using that bound we can conclude the systemic-risk (i.e., CAPM) inflation risk premium is bounded from above by $.03 \times 0.448 \cong 1.3\%$.

7 Policy Implications of the TIPS / Inflation Swaps Markets

Although it is not the primary focus of this paper, the issues discussed below do have one important public-policy implication. Specifically, the Federal Reserve Board of Governors have repeatedly referred to their benchmark of 2% for the average rate of inflation.

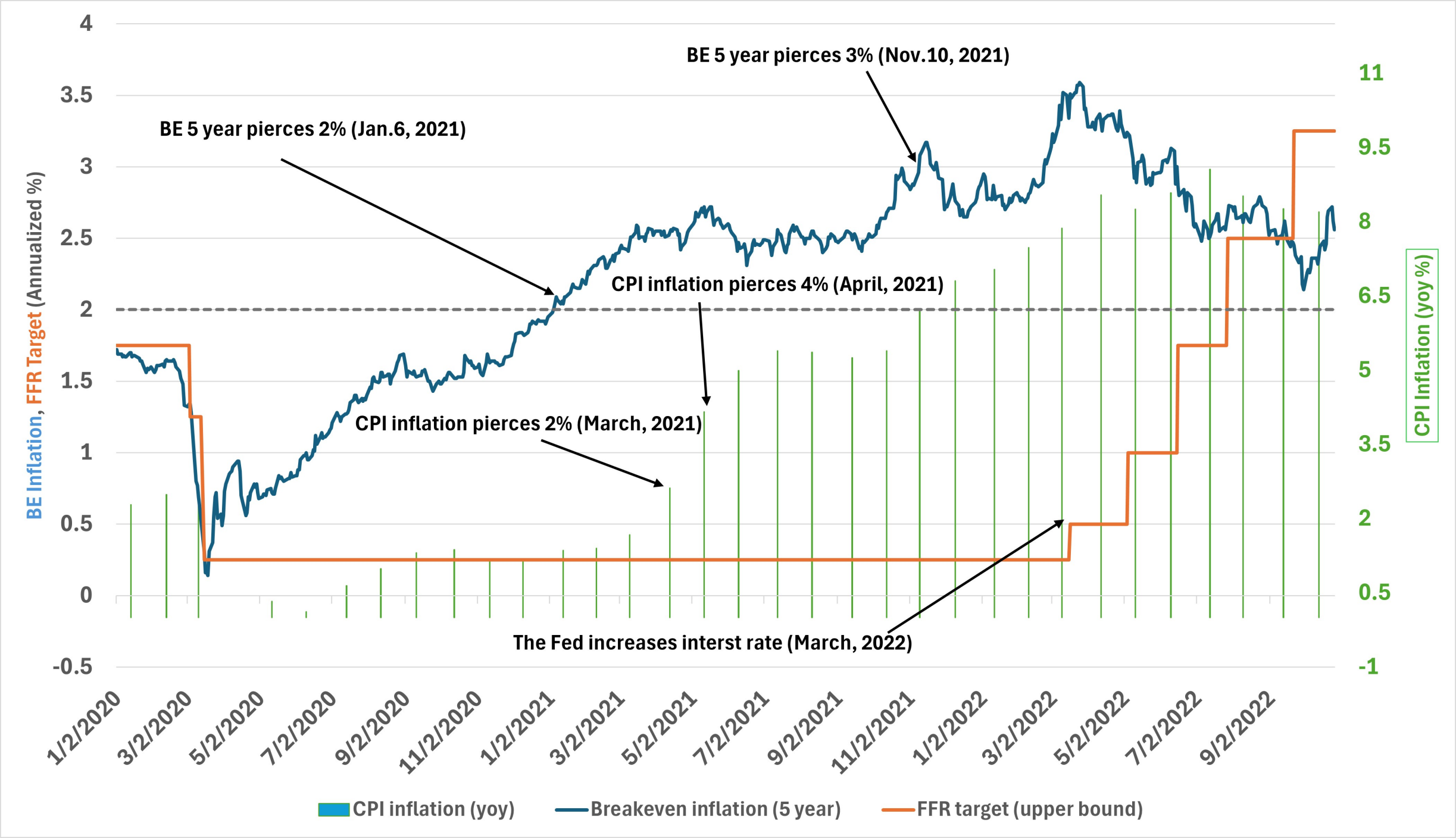
As is well-known, the Federal Reserve operates under the dual mandate of promoting maximum employment and maintaining stable prices, where the latter is typically interpreted as a 2% inflation rate, measured by the Personal Consump-

tion Expenditures (PCE) price index.[12] Thus, the first objective might be seen as giving the Federal Reserve some forbearance in terms of acting against a resurgent inflation.

The pandemic initially caused the five-yr. break-even rate of inflation to dip as low as 0.14% on March 18, 2020, that forward-looking rate of inflation rose consistently from that depth: It pierced 2% on Jan. 6, 2021, 2.5% on March 5, 2021 and 3% on Nov. 10, 2021. As Fig. 7 below demonstrates, this is not a matter of when it was appropriate to abandon the adjective "transitory." Rather, it demonstrates the Federal Reserve had ample notice well before they finally began increasing interest rates on March 16, 2022.

(PLACE FIG. 7 HERE)

[12]Even though PCE is what the Fed targets, the market perceives CPI to be a more relevant price measure. For example, in the *Bloomberg Economist Forecast* survey on macroeconomic releases, there is a much higher number of participants in the CPI release compared to the PCE inflation release. In fact, CPI is one of the measures with the highest number of survey participants.


BE 5 year pierces 3% (Nov.10, 2021)
BE 5 year pierces 2% (Jan.6, 2021)
CPI inflation pierces 4% (April, 2021)
CPI inflation pierces 2% (March, 2021)
The Fed increases interst rate (March, 2022)
BE Inflation, FFR Target (Annualized %)
CPI Inflation (yoy %)
4
3.5
3
2.5
2
1.5
1
0.5
0
-0.5
11
9.5
8
6.5
5
3.5
2
0.5
-1
1/2/2020
3/2/2020
5/2/2020
7/2/2020
9/2/2020
11/2/2020
1/2/2021
3/2/2021
5/2/2021
7/2/2021
9/2/2021
11/2/2021
1/2/2022
3/2/2022
5/2/2022
7/2/2022
9/2/2022
CPI inflation (yoy)
Breakeven inflation (5 year)
FFR target (upper bound)

8 Conclusions

In this paper, we have sought to address the sign and magnitude of the inflation risk premium, while accounting for liquidity effects and gauging the impact of inflation when measured against the systematic CAPM component.

Perhaps surprisingly, it turns out the inflation risk premium, defined as the excess of the market-based inflation above and beyond expected inflation, is negative, most clearly and robustly at the short horizon, both in accordance with unconditional tests, survey-based results and the application of the CAPM.

A simple CAPM framework proposed in this paper rationalizes the negative sign: because inflation is viewed pro-cyclical, nominal bonds work as a hedging asset in bad states and therefore command a negative premium. Three independent exercises converge at the negative premium at the near horizon, where the evidence is strongest: the sign-concurrence test, the predictability regression, and the close tracking of the CAPM-based risk-adjusted expectations to the survey. The same framework implies a tight upper bound on the premium of roughly 1.3 percent.

In terms of liquidity effects, we have documented the maturing of the TIPS market in the early 2000 – 2001 period, while a mild but persistent downward bias in break-evens lingers. However, our results also hold in the inflation-swap market, which is less subject to this distortion, suggesting our conclusions are not driven by TIPS-specific illiquidity.

An important consequence of this is that it behooves all consumers of this data, including policy makers, investors, and businesses to account for the presence of a negative premium as they use market-based measures of inflation in their planning and projections. In particular, we have documented how the Federal Reserve Board of Governors failed to discern the inflation signs clearly emitted by markets in the year prior to their March 2216, 2022 increase in the Federal discount rate.

A Literature Review

A.1 Overview of the U. S. Index-Linked Bonds

Subsequent to the issuance of the first TIPS security in Jan. 1997, researchers have weighed in with their analyses of this burgeoning market:

1. Sack and Elsasser (2004) describe the development of Treasury Inflation-Indexed Securities (TIIS) since 1997, noting that initially, TIIS yields were surprisingly high compared to nominal Treasuries due to factors like lower liquidity and investor unfamiliarity. However, market conditions and valuation of TIIS improved as liquidity and investor participation increased over time.

2. Campbell, Shiller and Viceira (2009), which examines the U. S. and UK inflation-indexed bond markets, highlighting a significant drop in long-term real interest rates from the 1990s until 2008, followed by a sharp rise during the 2008 financial crisis. It also notes that break-even inflation rates were stable from 2003 until late 2008, when they experienced dramatic declines.

A.2 Measuring the Inflation Risk Premium

The issue of measuring the inflation risk premium has been addressed in the literature:

1. Durham (2006) "decomposes nominal Treasury yields into expected real rates, expected inflation rates, real risk premiums, and inflation risk premiums by separately calibrating a three-factor affine term structure model to the nominal Treasury and TIPS yield curves."

2. Kupfer (2018) presents a meta-analysis and methodological road-map for anyone trying to understand or estimate inflation risk premia using bond markets.

3. Chen, Liu and Cheng (2010) estimate "the term structure of inflation risk premia implied from the TIPS market, [finding that] from the period of January 1998 through October 2007 the expected inflation rate is very stable and the inflation risk premia exhibit a positive term structure."

4. Pericoli (2012) shows that "forward inflation risk premia become sizable around the start of the late-2000s financial crisis and considerably increase just before the adoption of the first unconventional monetary policy measures in March 2009."

5. Sarte (2012) uses "a simple consumption-based asset pricing model to investigate the significance of the inflation risk premium [and finds] that historical estimates of the inflation risk premium are inconsequential."

6. Hördahl and Tristani (2012) consider European results: "Our results indicate that the inflation risk premium on euro area 10-year nominal yields was approximately equal to 20 basis points on average over the 1999-2007 period."

7. Bauer and McCarthy (2015) find "that market-based inflation forecasts generally are no more accurate than surveys of professional forecasters or simple forecast rules. This suggests that financial markets can provide little additional useful forward-looking information about inflation."

8. Casiraghi and Miccoli (2019) claim the "rate of swap contracts linked to inflation can be a poor measure of inflation expectations, as it incorporates time-varying risk premia. Our results show that premia are negatively related to the business cycle and the volatility of the stock market, increase with the maturity of the contract and are on average lower in the US than in the euro area."

9. Binder and Kamdar (2022) show that expected and realized inflation are strongly correlated over long samples, less so over short intervals.

A.3 The Impact of Liquidity Risk

1. Shen and Corning (2001) "yield difference has not provided a good measure of market inflation expectations because of the large and variable liquidity premium on TIPS. Still, the yield difference may become a better measure of market inflation expectations as liquidity conditions in the two kinds of Treasury markets move closer in the future."

2. Shen (2006) points out that "attributing changes in yield spreads to changes in market inflation expectations and ignoring the liquidity risk premium could lead to overstated inflation expectations liquidity risk premia, which have likely declined in recent years."

3. Christensen and Gillan (2011) use information from inflation swaps to "provide a range for the possible liquidity premium in TIPS, which in turn suggests a range for estimates of inflation expectations well below the Survey of Professional Forecasters inflation forecast."

4. Zeng (2013) "decomposes the break-even inflation rates derived from inflation-indexed bonds into inflation risk premia, liquidity risk premia, and inflation expectations. The results indicate that the sharp declines in the 10-year and 5-year break-even inflation rates in 2009 reflect a substantial increase in liquidity risk rather than a decrease in inflation expectations. The model-implied inflation expectations show better forecast performance for the average annual inflation rates than raw break-even inflation rates, the Survey of Professional Forecasters, and the Surveys of Consumers inflation forecasts."

5. Fleckenstein, Longstaff and Lustig (2014) contrast the price of a Treasury bond and an inflation-swapped TIPS issue, "exactly replicating the cash flows of Treasury bond [and show that the two] can differ by more than $20 per $100 notional. Treasury bonds almost always overvalued relative to TIPS."

6. Driessen, Nijman and Simon (2017) "show that the level of liquidity affects TIPS, whereas swap yields include a liquidity risk premium. These results are based on a model with a systematic liquidity risk factor and asset-specific liquidity characteristics. We show that these liquidity (risk) premiums explain a substantial part of the TIPS underpricing."

7. Andreasen, Christensen and Riddell (2021) use "a novel model to identify liquidity risk from individual TIPS prices by accounting for the tendency that TIPS go into buy-and-hold investors' portfolios. We find a sizable and countercyclical TIPS liquidity premium. Accounting for liquidity risk also improves the model's ability to forecast inflation and match surveys of inflation expectations."

A.4 Relating the Inflation Risk Premium to the CAPM's Systematic Risk

1. Dokko and Edelstein (1991) "discuss how inflation risks are capitalized into asset values and returns."

2. Wei and Wong (1992) "examine the relationship between stock returns and inflation across 19 industry sectors during the pre- and post-World War II periods. The results in the postwar period can explain the spurious negative relationship between stock returns and expected inflation in all industries, [except] non-natural-resource industries. The expected inflation/returns sensitivity appears to be positively related to the level of real assets, but negatively related to debt ratio during the postwar period."

3. Balsam, Kandel and Levy (1998) "quantify, with the use of a consumption-based CAPM, the risk premiums inherent in the Israeli market for index-linked and non-index-linked bonds. It is shown that the inflation risk premium term and the indexation-lag risk premium term depend heavily and positively on the degree of relative risk aversion, and that the latter is inconsequential. As a result, we claim that the bias caused due to overlooking both these risk premiums in the computation of inflation expectations depends on assumptions regarding the degree of relative risk aversion."

4. Campbell, Pflueger and Viceira (2020) find that the correlation between inflation and the output gap shifted from negative to positive in 2001. It also emphasizes the role of risk premia in amplifying the resulting changes in the comovement of bond and stock returns.

5. Cieslak and Pflueger (2023) explores the relationship between inflation and the business cycle, showing how inflation trends align with economic fluctuations and how stocks correlate with Treasury bonds.

B The Analytics of the Liquidity Premium

To illustrate the intuition behind this test, denote y_t^N the observed yield on nominal bond and r_t the real interest rate.. Based on the Fisher equation and assuming the

existence of inflation risk premium, we have

$$y_t^N = r_t + E_t(\pi) + \text{IRP}_t$$

IRP_t and LRP_t represent inflation and liquidity risk premia, respectively. Define y_t^R the yield observed from the inflation-indexed bonds (e.g. TIPS). Assuming the presence of liquidity risk premium:

$$y_t^R = r_t + \text{LRP}_t$$

The augmented Fisher equation can be rewritten as

$$y_t^N = y_t^R + E_t(\pi) + \text{IRP}_t - \text{LRP}_t$$

Equivalently, in the first difference:

$$\Delta y_t^N = \Delta y_t^R + \Delta E_t(\pi) + \Delta\,\text{IRP}_t - \Delta\,\text{LRP}_t$$

Consider now the slope coefficient of regression (4), namely, the regression of Δy_t^N on Δy_t^R. Here, we have

$$\text{Cov}(\Delta y_t^N, \Delta y_t^R) = \text{Cov}(\Delta y_t^N, \Delta y_t^N - \Delta E_t(\pi) - \Delta\,\text{IRP}_t + \Delta\,\text{LRP}_t)$$

Then, the regression slope is

$$\begin{aligned}\beta &= \frac{\text{Cov}(\Delta y_t^N, \Delta y_t^R)}{\text{VAR}(\Delta y_t^N)} \\ &= 1 - \frac{\text{Cov}(\Delta y_t^N, \Delta E_t(\pi) + \Delta\,\text{IRP}_t)}{\text{VAR}(\Delta y_t^N)} + \frac{\text{Cov}(\Delta y_t^N, \Delta\,\text{LRP}_t)}{\text{VAR}(\Delta y_t^N)} \qquad \text{(B1)}\end{aligned}$$

The first covariance term on the RHS of (B1) should be positive, as it indicates the covariance between changes in nominal yields and the sum of changes in inflation expectations and inflation risk premia $\left(\text{Cov}\left[\Delta y_t^N,\ \Delta E_t(\pi) + \Delta\,\text{IRP}_t\right]\right)$. This is due to the relationship that an increase in either inflation expectation or inflation risk premia will induce a positive change in nominal yield. Moreover, the second term, which represents the covariance between the changes in the nominal yield and the changes in the liquidity risk premia $\left[\text{Cov}\left(\Delta y_t^N,\ \Delta\text{LRP}_t\right)\right]$, should be negative. Holding other terms fixed, a positive change in LRP will decrease the observed real interest rate, and hence decrease the nominal interest rate as well.

In other words, the formula suggests that the slope coefficient should be just below unity. Under a mature market, the first covariance should be stable and only exhibit mild fluctuations as the nominal and real yields should behave in a way consistent with the Fisher equation augmented with IRP. The second covariance should diminish as the effect of liquidity premia on the nominal yield should weaken as the market matures. Thus, the regression test enables us to identify the timing of the maturity of the market by finding a period where the slope coefficient stabilizes at some point below unity.

C Derivation of the Inflation Risk Premium under the CAPM

In the body of the text, we derive the result (10):

$$E\left(\text{Inflation}\right) \;=\; y_{\text{TREAS}} - y_{\text{TIPS}} + \text{Corr}\left(\text{Inflation},\, r_m\right) \text{Std}\left(\text{Inflation}\right) \text{Sharpe Ratio}$$

Furthermore, note that

$$\text{Cov}\left(r_{\text{TIPS}},\, r_m\right) = \text{Cov}\left(y_{\text{TIPS}} + \frac{d\pi}{\pi},\, r_m\right),$$

If this holds conditional at each time t:

$$\text{Cov}_t\left(r_{\text{TIPS},t+1},\, r_{m,t+1}\right) = \text{Cov}_t\left(y_{\text{TIPS,t+1}} + \frac{d\pi_{t+1}}{\pi_t},\, r_m\right),$$

adding and subtracting the time t inflation rate $\left(\frac{d\pi_t}{\pi_{t-1}}\right)$ gives

$$\text{Cov}_t\left(r_{\text{TIPS},t+1},\, r_{m,t+1}\right) = \text{Cov}_t\left(y_{\text{TIPS,t+1}} + \frac{d\pi_{t+1}}{\pi_t} + \frac{d\pi_t}{\pi_{t-1}} - \frac{d\pi_t}{\pi_{t-1}},\, r_m\right).$$

This can be rewritten as

$$\text{Cov}_t\left(r_{\text{TIPS},t+1},\, r_{m,t+1}\right) = \text{Cov}_t\left(y_{\text{TIPS,t+1}} + \Delta\,\text{Inflation} + \frac{d\pi_t}{\pi_{t-1}},\, r_m\right),$$

where Δ Inflation $= \dfrac{d\pi_{t+1}}{\pi_t} - \dfrac{d\pi_t}{\pi_{t-1}}$. Note that at time t, $d\pi_t/\pi_{t-1}$ is known and constant. Thus, $\text{Cov}_t\left(\dfrac{d\pi_t}{\pi_{t-1}}, r_m\right) = 0$. With the original assumption that $\text{Cov}_t\left(y_{\text{TIPS},t+1}, r_m\right)$ is zero, the covariance results in

$$\text{Cov}_t\left(r_{\text{TIPS},t+1},\, r_{m,t+1}\right) = \text{Cov}_t\left(\Delta\,\text{Inflation}, r_m\right),$$

Thus, conditional at time t, we have

$$\begin{aligned} E_t\left(\text{Inflation}\right) &= \text{Break-Even Inflation}_t + \text{Corr}_t\left(\text{Inflation},\, r_m\right) \text{Std}_t\left(\text{Inflation}\right) \text{SR} \\ &= \text{Break-Even Inflation}_t + \text{Corr}_t\left(\Delta\,\text{Inflation},\, r_m\right) \text{Std}_t\left(\text{Inflation}\right) \text{SR} \end{aligned} \tag{C1}$$

D Empirical Evidence in Support of $\mathbf{Cov}\left(y_{\mathbf{TIPS}},\, r_m\right) = 0$ Empirical Evidence in Support of $E\left(\Delta y_{\mathbf{TIPS}}\right) = 0$

The first hypothesis being empirically tested here is that the real yield is uncorrelated with the (nominal) rate of return on the market portfolio. To do this, we run the following regression analysis:

$$y_{\text{5-yr. TIPS}} = \alpha + \beta\, r_m$$

The second hypothesis examines whether empirically $E\left(\Delta y_{\text{TIPS}}\right) = 0$.

Table D1 — Regressing 5-Yr. Real Yields on Nominal Market Returns

Dates:	8/23/23 – 8/23/25
Data:	Weekly
No. of Observations:	104
R^2	0

	Intercept	Slope
Coefficient	1.83	0
t-stat		−0.019

The t-stat for $E\left(\Delta y_{\text{TIPS}}\right) = 0$ is −0.694.

Note: The regression's slope was indistinguishable from zero to four decimal points.

E Predictability Regression: Break-even and the Liquidity Confound

For completeness, we report the break-even counterpart of the inflation-swap regression, $\text{IRP}_{2,h} = a + b\,\beta_{\Delta BE,h}$. The break-even beta is a statistically insignificant, and often wrong-signed, predictor at every horizon.

Table E1 — Predictability Regression: Break-even

Horizon	n	conc.	slope b	t	p	R^2
1yr	53	0.755	0.004	0.06	0.952	0.000
5yr	53	0.849	−0.012	−0.36	0.717	0.002
10yr	53	0.849	0.000	0.00	0.999	0.000

Notes:

1. The estimated regression result from $\text{IRP}_{2,h} = a + b\,\beta_{\Delta BE,h}$.
2. The regressor is standardized by its own standard deviation. So b is the pp change in the IRP per one standard deviation of beta.
3. "conc." is the sign-concurrence ratio, which uses the same observations as the regression.
4. t and p are t-statistics and p-values computed using the Newey-West standard errors.

This is what we would expect once the break-even is written as the swap less the TIPS liquidity premium (for simplicity, we abstract from the difference in the coupon structure), $\text{BE}_t = \text{IS}_t - \text{LRP}_t$, so that the market betas decompose as

$$\beta_{\Delta\text{BE},\,r_m} = \beta_{\Delta\text{IS},\,r_m} - \beta_{\Delta\text{LRP},\,r_m}.$$

The swap beta measures the inflation–market comovement the CAPM prices. The break-even beta equals it less a liquidity-premium beta. Because that liquidity-premium component is unrelated to the inflation risk premium, it contaminates the break-even beta as a regressor: it can shift the slope and even flip its sign, so the break-even predictability regression need not recover the priced inflation–market relationship and its coefficient is economically uninterpretable. The inflation swap, which carries no TIPS liquidity leg, is the clean regressor.

F Robustness to the Rolling-window Size

To check for the robustness of the results to the size of the rolling window, we recompute the key results at windows of 30 and 60; the construction is otherwise identical, the window-45 entries reproduce the baselines, and all results are stable.

Table F1 — Sign-Concurrence Ratios by Window Size

	break-even			inflation swap		
Window	1yr	5yr	10yr	1yr	5yr	10yr
30	0.717	0.868	0.849	0.887	0.698	0.491
45	0.755	0.849	0.849	0.887	0.660	0.434
60	0.698	0.849	0.830	0.868	0.660	0.453

Notes:

1. Each cell is the unconditional sign-concurrence ratio between the survey IRP and the CAPM beta, computed at rolling windows of 30, 45 and 60 observations.
2. The window-45 row reproduces the baseline sign-concurrence ratios (Table 2, Panel A).

Next, we present the volatility share recomputed under different window sizes.

Table F2 — Volatility Share by Window Size

	BE 5yr	BE 10yr	IS 5yr	IS 10yr
window 30	57.3%	65.3%	64.5%	79.9%
window 45	57.2%	62.6%	64.8%	75.3%
window 60	55.5%	60.7%	60.7%	71.6%

Note:

1. Each cell is the volatility share of the 1yr $\rightarrow h$ change in the CAPM premium. The correlation share is the complement, and the two sum to 100%.

Lastly, we present the predictability regression results that use betas measured under different window sizes.

Table F3 — Predictability-Regression Slopes by Window Size

Series	window 30	window 45	window 60
IS 1yr	−0.197 (−2.58)	−0.158 (−2.37)	−0.099 (−2.39)
BE 1yr	−0.038 (−0.44)	0.004 (0.06)	0.048 (0.64)
IS 5yr	−0.057 (−1.54)	−0.010 (−0.29)	−0.019 (−0.56)
BE 5yr	−0.043 (−1.13)	−0.012 (−0.36)	−0.016 (−0.53)
IS 10yr	−0.052 (−1.43)	−0.044 (−1.18)	−0.042 (−1.12)
BE 10yr	−0.031 (−1.07)	0.000 (0.00)	−0.008 (−0.22)

Notes:

1. Each cell is the standardized slope b with its Newey-West t-statistic in parentheses, at rolling windows of 30, 45 and 60.
2. The window-45 column matches the predictability regressions in the main text and the appendix.
3. The one-year inflation-swap slope is negative and statistically significant at every window.

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